Measuring Competition and Cooperation in LLM Bargaining: An Empirical Meta-Game Analysis

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Abstract

We conduct an empirical game-theoretic analysis of large language models (LLMs) negotiating to split a collection of subjectively valued indivisible items, as a case study of assessing advanced AI in mixed-motive settings. Bargaining agents built on proprietary LLMs represent *meta-strategies*, mapping prompts describing the game to negotiation policies. A graduated series of prompting levels steer agents away from blatant negotiation errors. For a variety of bargaining scenarios, we estimate *empirical meta-games* over meta-strategies that include LLM agents, three heuristic strategies representing extreme negotiation attitudes, two reinforcement learning-derived policies, and an agent based on common bargaining heuristics. We evaluate agents at meta-game equilibria in terms of individual effectiveness, social welfare, and fairness, using bootstrap methods to quantify uncertainty. Our analysis reveals a positive association between individual effectiveness and social metrics across models, with performance varying systematically across providers, architectures, and prompting levels. Behavioral analyses further illuminate why certain models excel or fail in specific bargaining regimes and uncover distinct qualitative patterns across OpenAI, Anthropic, and Google models.

1 Introduction

With rapid advances in large language model (LLM) technology and its growing deployment in diverse application domains, there is a pressing need for principles and tools to assess its beneficence and safety. LLMs are typically evaluated with respect to static benchmarks in terms of the quality of responses to given inputs. However, LLM agents are increasingly more likely to operate in environments where they must interact with other autonomous AI-powered entities. In such multiagent environments, performance inherently depends on the behavior of other agents. A central principle of our research endeavor is that a good evaluation approach for LLM agents, and advanced AI agents more generally, must carefully consider its assumptions about other-agent context and the sensitivity of results to these strategic assumptions.

For our case study of a controlled multiagent evaluation framework for sophisticated AI agents, we focus on the important strategic domain of sequential negotiation (i.e., ***bargaining***) under incomplete information. The concrete variant of negotiation we utilize is an extension of a scenario employed in many previous AI studies (e.g., by Lewis et al. [2017]): finite-horizon, alternating-offer bargaining with the goal of partitioning a set of indivisible items between two parties with private valuations and exogenous fallback options (see Section 3.1 for details). Negotiation appears in a wide range of real-world situations, from interpersonal to international relations, and has a long history of being studied across disciplines. Each party in a negotiation has its own interests, but can succeed only by finding deals that are acceptable to all parties. Thus, negotiation is a canonical example of ***mixed-motive*** settings, simultaneously involving *competition* over the potential gains from a deal and *cooperative* search for mutually beneficial agreements. This attribute makes negotiation a natural, yet challenging, scenario for assessing the emergent properties of AI in multiagent contexts.

Having chosen this task environment, we must carefully consider AI negotiator desiderata as the next step in constructing our evaluation framework. The obvious purpose of these AI agents is to be effective in identifying and reaching deals for mutual benefit. Intuitively, this means that they should be able to understand the respective principals' circumstances and objectives on the one hand, and behave in a transparent, trustworthy manner on the other; for example, they should not misrepresent facts or express insincere commitments. Moreover, their separate training should not produce unanticipated negative outcomes from mutual interactions. Although it is unrealistic to expect an assessment framework to cover all possible nuances of AI negotiator behavior, it is clear that it must involve multiple criteria, both quantitative and qualitative, formulated from both individual success-oriented and prosocial perspectives.

Moreover, a good evaluation approach must also account for various forms of uncertainty that can arise in the task environment under consideration.

We address the following broad research question within this bargaining game environment:

> **RQ.** *What is the interplay between individual (strategic) effectiveness and socially desirable behavior exhibited by advanced AI negotiators?*

We make the following methodological contributions in pursuit of this question:

- **Hierarchical prompting framework.** We construct a structured, incremental prompting framework to adapt select proprietary LLMs to the bargaining task under consideration. Our objective here is not to optimize a workflow incorporating an LLM for our specific task—instead our framework enables us to systematically analyze models' inherent competence in our bargaining task when provided with graduated scaffolding.

- **Multi-criteria assessment.** We define and employ a slate of metrics to assess and compare LLM-based bargaining agents. As indicators of individual effectiveness, we use game-theoretic regret and best-response graphs; to measure social desirability, we apply standard welfare measures and an envy-based fairness criterion to negotiation outcomes.

- **Baselines for robustness analysis.** We include in the mix two baseline strategies derived via reinforcement learning (RL) and three representing extreme behaviors: agreeing immediately, walking away immediately, and rigidly insisting on a selfish deal. Additionally, we include a baseline concession-schedule strategy, modeled after examples employed in negotiation agent competitions [Mohammad et al., 2020].

- **Empirical meta-game evaluation.** We systematically collect data from head-to-head bargaining between our agents under carefully chosen parameter configurations, use the data to estimate empirical meta-game models [Li and Wellman, 2024], and compute their equilibria, which drives further analysis. We characterize uncertainty through statistical bootstrapping techniques. By using distributions of equilibrium-adjusted measurements rather than point estimates, we produce robust and context-rich results on LLM performance in a stochastic strategic environment. We focus our analysis at the meta-strategy level, identifying patterns among individual models cast as meta-strategies, and we also provide a higher-level analysis by aggregating results for all models from the same vendor to uncover patterns between model providers.

Our analysis reveals a consistent positive correlation rather than a tradeoff between individual effectiveness and social metrics across bargaining scenarios. We find the same models tend to lead rankings across metrics. Moreover, some models end up being exploited by the rigid heuristic strategy, exposing an interesting form of vulnerability.

2 Background and Related Work

In the context of single-agent evaluation, Jordan et al. [2020] highlight the importance of robust benchmarks for evaluating agent capabilities. These are likewise essential in multiagent settings and further complicated by the need to address strategic interactions. Recent "reevaluations" [Balduzzi et al., 2018, Liu et al., 2025] underscore the inadequacy of coarse metrics like win rates in capturing nuanced agent behaviors. Work in evaluating cooperation [Papoudakis et al., 2021, Sinha and Wellman, 2019] emphasizes the need for welfare-centric evaluations.

Outcomes in multiagent environments inherently depend on the joint behavior of all agents. A particular agent's performance, therefore, hinges on which agents it plays against: the ***strategic context***. Balduzzi et al. [2018] argue that the most relevant strategic contexts are where other agents are approximately rational, and advocate evaluation with respect to a game-theoretic equilibrium, which they term ***Nash-averaging***. This is equivalent to the concept of ***NE-regret***, employed by Jordan et al. [2007] to evaluate agents in a research competition.

A major inspiration for our framework is the meta-game evaluation approach introduced by Li and Wellman [2024], which uses empirical game-theoretic analysis (EGTA) [Wellman et al., 2025] to construct a game model over agent meta-strategies defined by multiagent RL algorithms. This work also demonstrated the use of statis-

tical bootstrapping techniques [Davison and Hinkley, 1997] to provide distributional assessments of game-theoretic properties [Wiedenbeck et al., 2014].

The multi-item bargaining scenario that we investigate is related to a formulation popularized within AI by Lewis et al. [2017] and employed subsequently by many researchers [Cao et al., 2018, He et al., 2018, Lanctot et al., 2019, Chawla et al., 2023, Hua et al., 2024b, Li and Wellman, 2024, Li et al., 2025, Konicki et al., 2025]. The centralized version of this scenario, where a non-strategic arbiter needs to allocate a set of indivisible resources to two or more agents fairly and efficiently, given access to agents' subjective valuations over those resources, is a long-standing area of active research [Amanatidis et al., 2023]. A key fairness property in this context is envy-freeness up to one item (EF1), implicit in the work of Lipton et al. [2004] and explicitly formalized by Budish [2011]. Caragiannis et al. [2019] connected the envy-based fairness paradigm with welfare notions by proving that, for additive valuations, an allocation maximizing Nash welfare must satisfy the EF1 property; however, an approximation to the maximum Nash welfare is not known to provide any guarantees on envy relaxation [Amanatidis et al., 2023]. This body of theoretical work informs our evaluation criteria.

There is another relevant line of theoretical work on the *distributed* version of the indivisible resource allocation problem [Endriss et al., 2006, Ramezani and Endriss, 2010, Chevaleyre et al., 2017]. It characterizes local, boundedly rational agent behavior that results in the convergence of unbounded-horizon structured negotiations to an allocation with desirable welfare and fairness properties.

With the advent of GPT-class LLMs, several large-scale assessments of the strategic reasoning and decision-making capabilities of LLM-based agents have been conducted on various multiagent/interactive testbeds. These include the simulated auction environment AUCARENA [Chen et al., 2023], classical game-theoretic settings [Fan et al., 2024, Duan et al., 2024], GTBENCH [Duan et al., 2024] that features widely studied board and card games, and AGENTBENCH that includes game-grounded environments [Liu et al., 2024]. Hua et al. [2024a] focus on prompt engineering to design LLM workflows for strategic decision-making and evaluate the outcomes of applying these workflows to a suite of game-theoretic environments, including a bargaining setup similar to ours, in terms of efficiency and fairness criteria (e.g., Pareto optimality and envy-freeness). Abdelnabi et al. [2024] propose an LLM negotiation benchmark where multiple agents aim to reach deals over a set of issues. NegotiationArena [Bianchi et al., 2024] provides a complementary framework, instantiating several bargaining and trading games and showing that simple behavioral tactics can substantially shift LLM negotiation outcomes. Many studies such as that by Murashige and Ito [2025] have probed into LLM performance in variants of the ultimatum game, which can be considered a special one-round restriction of our bargaining scenario. Cookson et al. [2026] examined the fairness properties of LLMs as arbiters of the centralized indivisible item allocation problem depending on contextual and prompt framing.

Parallel social science research employs LLMs to simulate human-like participants, identifying biases and interaction heuristics [Horton, 2023, Argyle et al., 2023, Aher et al., 2023]. These findings emphasize the need to audit LLM interactions beyond competitiveness, incorporating fairness and welfare considerations. Emergent cooperation among multiagent LLM systems has also been explored, for instance by Piatti

et al. [2024] through societal simulations.

3 Technical Preliminaries

Let $\mathbb{Z}_{>0}$ (respectively, $\mathbb{Z}_{\geq 0}$) denote the set of all positive (respectively, non-negative) integers, and $\mathtt{U}(S)$ the uniform discrete probability distribution with support $S \in \mathbb{Z}_{>0}$.

3.1 Bargaining Game

We study an alternating-offer bargaining scenario between two parties, labeled player 1 and player 2, over a public pool of indivisible items. There are $T \in \mathbb{Z}_{>0}$ distinct item types and q_t items of type $t \in \{1, \ldots, T\}$ in the pool, defining a ***quantity vector*** $q = (q_1, \ldots, q_T) \in \mathbb{Z}_{>0}^T$. The goal of bargaining is to determine an ***allocation*** $a = (a_1, a_2) \in \mathbb{Z}_{\geq 0}^T \times \mathbb{Z}_{\geq 0}^T$ where a_i, called a bundle, is a vector specifying the number of items of each type assigned to player $i \in \{1, 2\}$. An allocation must induce a 2-way partition over the item pool (no item is left unassigned). Thus, $a_1 + a_2 = q$.

All items of the same type have the same valuation from a given player's perspective. The parameter $V \in \mathbb{Z}_{>0}$ is the largest valuation that any player could have for any item type. For each player $i \in \{1, 2\}$ and each type $t \in \{1, \ldots, T\}$, we independently draw $v_{i,t} \sim \mathtt{U}(\{1, \ldots, V\})$, defining a ***private valuation vector*** $v_i = (v_{i,1}, \ldots, v_{i,T})$. Bundle valuations are additive: player i values bundle a_i at the scalar product $v_i \cdot a_i$.

Additionally, each player has an ***outside option*** $b_i \in \mathbb{Z}_{>0}$, called the ***best alternative to a negotiated agreement*** (BATNA) [Fisher et al., 2011, Ch. 6], which represents the value the player obtains as recourse if the parties fail to make a deal. We draw $b_i \sim \mathtt{U}(\{1, \ldots, v_i \cdot q\})$, which ensures that each BATNA is positive but does not exceed the relevant player's valuation for the full item pool. These bounds preclude strategically uninteresting edge cases. Each player observes its own valuation vector and BATNA, but not the other's. Without positive-valued private BATNAs, the negotiation scenario could degenerate into an ultimatum game [Harsanyi, 1961].

Negotiation unfolds over a maximum of R ***rounds***. Player 1 takes the first action in each round. On its turn, a player has the following actions to choose from: make an OFFER in the form of an allocation, ACCEPT the latest offer from the other, or WALK away. Negotiation ends on an ACCEPT or WALK action, or after R rounds; a final-round (counter)offer by player 2 is thus equivalent to WALK. ACCEPT results in an allocation a corresponding to the accepted offer, and each player i realizes a value $w_i = v_i \cdot a_i$. Otherwise, negotiation fails, and each player i realizes its BATNA, $w_i = b_i$. Values are exponentially discounted over rounds at a rate $\gamma \in (0, 1]$. Hence, if bargaining ends in round r, each player's ***payoff*** is given by $u_i = w_i \gamma^{r-1}$.

A ***bargaining game*** is defined by setting distributions of the parameters T, q, V, γ, R. Thus, in general, a game represents a class of bargaining scenarios with varying but controlled parameter configurations. For a particular game, one or more of these parameters may be constant.

3.2 Strategies and Utilities

Let $-i$ denote the player other than $i \in \{1, 2\}$. We denote by Π a set of ***pure strategies*** or policies such that $\pi_i \in \Pi$ assigns to player i a possibly stochastic choice over its available actions at each of its turns. In our study, a (pure) negotiation strategy may be implemented by an LLM, an RL-derived policy, or a handcrafted heuristic, some of which produce stochastic outputs (Section 4). A ***mixed strategy*** $\sigma_i \in \Delta(\Pi)$ is a probability distribution over pure strategies. Player i's ***utility*** for playing π_i against other-player strategy π_{-i} is denoted by $u_i(\pi)$, with $\pi = (\pi_i, \pi_{-i})$, and defined as the expectation of the player's payoff with respect to the generating distributions of the item pool, valuations, and BATNAs, as well as any stochasticity of actions induced by the pure strategies. For a mixed-strategy ***profile*** $\sigma = (\sigma_i, \sigma_{-i})$, player i's utility is defined as $u_i(\sigma) = \mathbb{E}_{\pi \sim \sigma}[u_i(\pi)]$. For one setting of T, q, V, γ, R and a realized sample of (v_1, v_2, b_1, b_2), assigning one strategy from Π to each player and letting negotiation rounds unfold constitutes one bargaining ***instance***.

3.3 Equilibrium and Regret

Strategy profile σ^* is a ***Nash equilibrium*** (NE) if no player can profitably deviate unilaterally, that is, for each i, $u_i(\sigma^*) \geq u_i(\sigma_i, \sigma^*_{-i})$ for any $\sigma_i \in \Delta(\Pi)$. A two-player game is ***symmetric*** if players' realized utilities are swapped if they swap strategies, hence the game can be characterized by a single utility function $u(\cdot, \cdot)$ where the first input is the strategy of the player under consideration, and second is that of the other player.[1] For a symmetric two-player game, we define the ***regret*** of a pure strategy π with respect to an NE σ^* as $\mathrm{Reg}_{\sigma^*}(\pi) \coloneqq u(\sigma^*) - u(\pi, \sigma^*_{-i})$.

For games with multiple NE, we focus on the ***maximum-entropy Nash equilibrium*** (MENE), defined as the NE σ^* that maximizes Shannon entropy $-\sigma^* \cdot \ln(\sigma^*)$. We approximate MENE using the mixed-integer linear programming approach of Li and Wellman [2024] with a tolerance of 10^{-6}.[2]

3.4 Welfare and Fairness Metrics

We consider three measures of an outcome's social welfare. The first is the widely used ***utilitarian welfare*** (UW) defined as the sum of players' payoffs. Maximizing ***Nash welfare***, defined as the geometric mean of payoffs, is a way to balance efficiency and fairness [Caragiannis et al., 2019]. We use two versions of Nash welfare, one based on the raw payoffs (NW) and the other on *advantages* defined as nonnegative surpluses

[1] Our bargaining game is symmetric if we randomize the assignments to roles 1 and 2. In our experiments we sample each profile ensuring equal numbers of the two role assignments.

[2] Liu et al. [2025] propose a criterion based on generalized Tsallis entropy [Tsallis, 1988] that is invariant to cloned strategies. Since our analysis does not include clones, we use Shannon entropy for equilibrium selection.

above respective BATNAs (NW$^+$):

$$\begin{aligned}
&\text{UW} \coloneqq u_1 + u_2;\\
&\text{NW} \coloneqq (u_1 u_2)^{\frac{1}{2}};\\
&\text{NW}^+ \coloneqq (u_1^+ u_2^+)^{\frac{1}{2}}, \quad u_i^+ = \max(0, u_i - b_i),\ i \in \{1, 2\}.
\end{aligned}$$

In our scenario, it is possible for a negotiation to end with an item allocation that gives at least one player a realized valuation below its BATNA, and we do observe such outcomes in our experiments. The raw-payoff version of Nash welfare makes no distinction among outcomes based on whether realized valuations are above or below BATNAs (or whether the outcomes correspond to successful deals, i.e., trajectories ending in an ACCEPT). The version using advantages, which truncates all below-BATNA realized valuations to zero, is aligned with Nash's classic bargaining solution [Nash, 1950]. We report both versions since they capture different perspectives on global impacts of bargaining strategies.

We also assess the fairness of negotiation outcomes using the envy-based paradigm. An allocation is ***envy-free*** if each player values its own allocated bundle at least as much as it values that of the other player. Since an envy-free allocation may not exist in general, we base our assessment on a relaxation that always exists: ***envy-freeness up to one item*** (EF1) [Lipton et al., 2004, Budish, 2011]. An allocation is EF1 if, for each player i, other-agent $-i$'s bundle is either empty or includes an item which, if hypothetically removed, would make i value its own bundle at least as much as $-i$'s diminished bundle. For our setting with additive valuations, this reduces to the condition $v_i \cdot a_{-i} - v_i \cdot a_i \leq \max_{t:a_{-i,t}>0} v_{i,t}$ for $i \in \{1, 2\}$ (see App. A).

We then define the ***EF1 frequency*** for a set of bargaining instances as the fraction of all instances ending in ACCEPT that produce EF1 allocations.

3.5 Empirical Meta-Game Analysis

Our evaluation methodology follows the empirical meta-game analysis approach of Li and Wellman [2024]. We simulate all combinations of a set of negotiation agents in various settings, and assemble the data in an empirical game model [Wellman et al., 2025]. We make this game symmetric by randomizing each agent into the roles of player 1 and player 2. From the game model, we derive empirical equilibria, and employ these equilibria as strategic contexts for evaluation of the slate of agents.

So that our context is uniquely defined, we focus on MENE in particular, and measure each agent's characteristics in terms of its performance against the MENE profile. To quantify uncertainty, we take a bootstrapping approach, constructing distributions over statistics of interest by resampling the empirical meta-game. For each game sample, we compute the associated MENE profile and assess the performance measures with respect to that MENE opponent mixture.

4 Agent Strategies

We tested agents employing three categories of strategies in our bargaining scenario: prompted LLMs, hand-coded heuristics, and RL-derived policies.

4.1 Large Language Model Agents

To cover a representative set of state-of-the-art LLMs (as of data collection time), we selected five models from three different providers. We left model parameters such as temperature at their default settings.

- **GPT-4o**, OpenAI's flagship multimodal model [Hurst et al., 2024], and **o3-mini**, its reasoning-oriented variant [OpenAI, 2025];[3]
- **Claude Sonnet 3.7** from Anthropic in both **standard** and **reasoning** modes [Anthropic, 2024];[4]
- **Gemini 2.0 Flash**, Google's lightweight baseline [Anil et al., 2023].[5]

We formulated a hierarchical prompting framework to adapt LLMs to our bargaining scenario. Based on preliminary explorations, we identified five qualitatively defined action patterns that a rational negotiation would never include, which we simply term ***mistakes***. An example is accepting an offer from the counterparty that one values less than one's own BATNA (see Appendix C for a complete list). We crafted seven nested levels of text prompting, which we label ***circles***. Circles 0–6 give progressively more elaborate yet high-level guidance, starting with a factual description of the bargaining scenario and moving on to explicitly emphasizing points that may circumvent the aforementioned mistakes. As bargaining rounds proceed, the players receive prompts presenting the bargaining history and action options (Appendix D.3). The prompt circles are as follows (see Appendix D for full descriptions):

- **Circle 0** (bare rules): specify items, private valuations, BATNA, available actions.
- **Circle 1** (objective specification): instruct the agent to maximize own realized valuation, considering its BATNA.
- **Circle 2**: show a worked numeric example of how to evaluate an offer [Wei et al., 2022].
- **Circle 3**: prescribe a step-by-step routine: assess, compare, then ACCEPT, WALK, or OFFER.
- **Circle 4**: list the five common mistakes we identified.
- **Circle 5**: supply quick numeric checks that guard against those mistakes.

[3] openai/gpt-4o-2024-08-06, openai/o3-mini-2025-01-31
[4] anthropic/claude-sonnet-3.7-2025-02-19
[5] google/gemini-flash-2.0-2024-12-15

- **Circle 6**: encourage the agent to infer its opponent's valuation from their offers and adapt accordingly.

This gives us a pool of $5 \times 7 = 35$ LLM-circle pairings. We call these pairings *LLM agents* and adopt the following naming convention for them: `<LLM>-c-<k>`, where

- `<LLM>` is the model identifier (`4o` for GPT-4o, `o3-mini` for OpenAI o3-mini, `sonnet-3.7` for Anthropic Claude 3.7 "Sonnet", `sonnet-3.7-r` for Anthropic Claude 3.7 "Sonnet Reasoning", `gem-2.0-f` for Gemini 2.0 Flash);
- `c` stands for "circle"; and
- `<k>` indexes the circle applied.

We narrowed our pool of LLM agents for the detailed analysis in Section 6 by conducting an experimental study that we detail in Appendix E. In summary, for each LLM agent, we ran 100 *self-play* instances (i.e., instances in which both players employ the same LLM agent as their strategies) from the BGL class as described in Section 5 and measured the frequency of each of our 5 mistakes in those instances. A salient observation from these experiments was that for some models, most notably Gemini 2.0 Flash, no single circle performed best across all mistakes. Ultimately, we used discretion to pick for each LLM the circle(s) that provided the most balanced performance in achieving low mistake frequencies. Our final set of LLM agents comprises `sonnet-3.7-c-5`, `sonnet-3.7-c-6`, `sonnet-3.7-r-c-0`, `gem-2.0-f-c-2`, `gem-2.0-f-c-5`, `4o-c-4`, `4o-c-5`, `4o-c-6`, and `o3-mini-c-0`.

4.2 Heuristic Strategies

We used four handcrafted heuristic strategies as baselines. The first three are interpretable, natural benchmarks that help us identify qualitative interaction patterns with the LLM agents:

- `tough` offers the opponent a single item of its least-valued type, and insists rigidly on this offer in each of its turns without ever picking ACCEPT or WALK. How other agents respond to this extremely unfavorable stance provides insights on their perception and vulnerability.
- `soft` (the antithesis of `tough`) accepts any offer received. When required to propose (i.e., as player 1 in round 1), generates an offer uniformly at random.
- `walk` simply walks away at the first opportunity. Negotiations involving `walk` thus end in round 1 with each player obtaining its BATNA.

Our final heuristic strategy is an adaptation of the **Aspiration** (`aspire`) negotiator from the NegMAS competition [Mohammad et al., 2020]. This strategy uses a polynomial concession schedule whose shape is controlled by a single parameter. We instantiate two common regimes: a *conceder* schedule (larger, earlier concessions), which we use in BG4, and a *Boulware* schedule (small early concessions that accelerate near the deadline), which we use in BG5 and BG6.[6] We provide further details in Appendix F.

[6] See Section 5 for descriptions of the games BG4, BG5, and BG6.

4.3 Reinforcement-Learned Strategies

We round out our strategy set with policies derived using two RL algorithms: ***Neural Fictitious Self-Play*** (`nfsp`) [Heinrich and Silver, 2016] and ***Regularized Nash Dynamics*** (`R-NaD`) [Perolat et al., 2021]. Appendices I and J provide our training hyperparameters for the two policies.

`nfsp` extends fictitious self-play [Heinrich et al., 2015] to extensive form games using neural-network function approximation. The algorithm maintains two distinct policies:

- a **best-response policy**, trained off-policy via Deep Q-Network (DQN) [Mnih et al., 2015] to exploit the current empirical average behavior of its opponents; and
- an **average policy**, derived through reservoir sampling [Vitter, 1985] of the agent's historical best-response policies, which approximates equilibrium play.

We use the average policy produced by `nfsp` as a baseline.

`R-NaD`, the algorithm underlying DeepMind's Stratego agent DeepNash [Perolat et al., 2022], employs follow-the-regularized-leader dynamics [Hennes et al., 2020]. It modifies the reward function of the game with a regularization term that prevents the cycling behavior typical of self-play dynamics [Mertikopoulos et al., 2018], which would otherwise require tracking an average policy to guarantee convergence. `R-NaD` maintains two components:

- a **regularized actor**, trained with model-free RL (policy-gradient or actor–critic) against self-play opponents while optimizing a reward modified by a divergence penalty relative to a reference policy; and
- an **anchor policy**, updated to reflect the equilibrium solution of the prior regularized game, serving as the anchor for the next stage of regularization.

By iteratively solving regularized games and updating the regularization anchor, `R-NaD` achieves last-iterate convergence to an NE of the original (non-regularized) game.

5 Experimental Setup

We consider two bargaining game classes, each parameterized by discount rate γ and number of rounds R, and with maximum value $V = 100$. For class BGL ('L' indicates large), we set $T = 5$ and sample $q_t \sim \texttt{Poisson}(4)$ for every t. For class BGS ('S' for small), $T = 3$ and $q = (7, 4, 1)$. Within each class, we focus on three games defined by (γ, R)-configurations: $(0.9, 3)$, $(0.98, 3)$, and $(0.98, 5)$. We call these games BG1, BG2, and BG3 respectively for BGL, and BG4, BG5, and BG6 respectively for BGS (see Appendix B for details).

Our first empirical meta-game analysis (`LLM-only`) focuses on the nine LLM agents as meta-strategies. For each game BGi, $i \in \{1, \ldots, 6\}$, we generated 10^3 bootstrapped empirical meta-game samples as follows. For each of our $9 \times 9 = 81$ pairs of LLM agents $(\mathcal{L}, \mathcal{L}')$, we sampled a cohort of $N_{(\mathcal{L},\mathcal{L}')} \approx 200$ settings of the

valuation functions and BATNAs (v_1, v_2, b_1, b_2), as described in Section 3.1, based on the relevant T and q. For each setting, we ran many bargaining instances, assigning the roles of players 1 and 2 to $\mathcal{L}$ and $\mathcal{L}'$ respectively for half of the cohort and flipping their roles in the remaining instances; for each instance, we recorded the realized payoffs (u_1, u_2) and whether the final accepted offer, if any, was an EF1 allocation. This collection of data underlies the empirical `LLM-only` game for $\mathrm{BG}i$. Note that by equalizing the assignments of agents to players 1 and 2 we can model the game as symmetric.

To analyze the empirical games we bootstrap distributions for the metrics of interest. A bootstrap game sample is constructed by resampling the data for all agent pairs. The data from agent pair $(\mathcal{L}, \mathcal{L}')$ is resampled $N_{(\mathcal{L},\mathcal{L}')}$ times with replacement. We take the resample-average payoffs as payoffs for the bootstrap sample, and correspondingly record welfare and EF1 frequency averages for the profile. After completing this resampling for all agent pairs, we compute the MENE of the bootstrap game sample, and evaluate a slate of metrics for each strategy with respect to the MENE (see Sections 3.3 and 3.4 for details). Our metrics are:

(1) regret Reg,

(2) expected normalized NW,

(3) expected normalized NW^+,

(4) expected normalized UW,

(5) expected EF1 frequency.

To normalize the welfare functions across games, we estimated the expected maximum welfare of each. Specifically, we generated millions of settings for each game under consideration, for each setting calculated the maximum possible utilitarian and Nash welfares over possible allocations (as well as the outcome in which each agent realizes its BATNA), and took the average of the maxima of each welfare function over all settings as the respective normalization constant.

Following Li and Wellman [2024], we construct for each game $\mathrm{BG}i$ an *empirical best-response graph* for a fine-grained view of strategic relationships. Each agent $\mathcal{L}$ is a node, and for every ordered pair $(\mathcal{L}, \mathcal{L}')$, we estimate an empirical best-response frequency $\hat{p}(\mathcal{L} \to \mathcal{L}')$: the proportion of the 10^3 bootstrapped $\mathrm{BG}i$ meta-games in which $\mathcal{L}'$ is the best response to $\mathcal{L}$ (i.e., achieves the highest average payoff among all agents when bargaining with $\mathcal{L}$). For better visualization, we draw a directed edge $\mathcal{L} \to \mathcal{L}'$ only when $\hat{p}(\mathcal{L} \to \mathcal{L}') \geq 0.01$. We make an edge dotted if $\hat{p} \in [0.01, 0.33]$, dashed if $\hat{p} \in (0.33, 0.66]$, and solid if $\hat{p} > 0.66$; moreover, a darker (and thicker) edge denotes a stronger response (i.e., a higher average payoff achieved by the best-response agent).

We produced aggregate results for the two game classes by constructing empirical meta-games from their respective empirical games. We define the meta-game $\mathrm{BGL0}$ as a uniform mixture over $\mathrm{BG1}$–3, where agents play their game-specific strategies. We bootstrap the empirical meta-game $\mathrm{BGL0}$ by bootstrapping each of $\mathrm{BG1}$–3, using the mean payoffs of the three constituent game samples for each agent pair. Each

bootstrap sample thus defines an MENE for the meta-game. We thus can compute average metrics at MENE and best-response graphs for the meta-game just as for the individual BGi. For our small game, we similarly define and analyze BGS0 based on BG4–6.

For our next analysis (`LLM+baselines`), we expand the scope to include our non-LLM baselines, as described above. For the smaller game (BGS) we included both heuristic and RL-based baselines, however training effective RL policies in BGL games proved too prohibitive. We further short-circuited computation involving the `walk` strategy by noticing that the result is BATNA for both players regardless of the other agent strategy, and this is realized at round 1 so γ and R are irrelevant. Hence, we generated one set of 3×10^5 single-round instances (in fact, just BATNA value pairs) for each of BGL and BGS that we plugged in for the `walk`-involved profiles for all games within the respective classes.

6 Results

We present results for meta-game BGS0 under `LLM-only` and `LLM+baselines`, with observations on strategy-level behavior, model-specific hallucinations, and provider-level aggregation. Tables in Figure 1 report bootstrap means for regret and welfare on BGS0, diagrams in that figure show the empirical best-response graphs. Detailed results for the specific games BG1 through BG6 plus meta-game BGL for the large game class are provided in Appendices H and R.

6.1 `LLM-only` Analysis

In `LLM-only`, the meta-strategies are exclusively LLM-based. The top table in Figure 1 reports performance on BGS0, and the top right panel shows the corresponding induced best-response graph.

Qualitatively, `o3-mini-c-0` is the strongest agent in the environment, while the remaining form a middle tier. Anthropic Sonnet variants, Gemini models, and the 4o family achieve broadly similar welfare and regret, with 4o slightly outperforming Gemini and Claude on most metrics. These differences are small compared to the gap between `o3-mini-c-0` and the rest.

For BGL0 (Appendix R) we find a similar ordering, with `o3-mini-c-0` again best, but a larger separation between the 4o family and the remaining models.

6.2 `LLM+baselines` Analysis

In `LLM+baselines` we add `R-NaD`, `aspire`, `nfsp`, `soft`, `walk`, and `tough`. The bottom panel of Figure 1 quantifies the resulting shift in strategic landscape. `o3-mini-c-0` remains among the top LLM agents. The two RL-derived meta-strategies, `R-NaD` and `nfsp`, exhibit notable performance: `R-NaD` is the only strong best response to `o3-mini-c-0`, and `nfsp` achieves especially high EF1 frequency. `R-NaD` is also the only meta-strategy with a significant self-edge. This manifests

BGS0 (**LLM-only**)

Agent	Reg↓	NW↑	NW$^+$↑	UW↑	EF1↑
sonnet-3.7-c-5	28.90±.50	76.33±.14	53.49±.14	83.29±.08	33.56±.27
sonnet-3.7-c-6	26.42±.55	74.86±.13	54.16±.13	83.15±.07	33.60±.30
sonnet-3.7-r-c-0	22.94±.49	78.96±.10	55.71±.15	84.98±.06	43.47±.33**
gem-2.0-f-c-2	24.06±.47	77.54±.14	54.19±.20	83.79±.07	38.37±.28
gem-2.0-f-c-5	22.20±.55	74.02±.20	49.46±.12	82.60±.08	37.21±.35
4o-c-4	17.43±.41**	79.29±.08**	53.45±.16	83.94±.04	37.21±.35
4o-c-5	22.88±.42	78.37±.13	56.09±.17**	84.41±.08	37.33±.33
4o-c-6	23.34±.42	78.25±.13	50.44±.11	84.10±.07	37.77±.29
o3-mini-c-0	**.06±.02****	**83.44±.10****	**68.19±.33****	**87.19±.07**	**56.56±.29****

BGS0 (**LLM+baselines**)

Agent	Reg↓	NW↑	NW$^+$↑	UW↑	EF1↑
sonnet-3.7-c-5	29.49±.39	74.42±.10	52.46±.19	82.32±.06	31.40±.19
sonnet-3.7-c-6	15.45±.47	74.51±.09	54.72±.20	82.88±.05	31.72±.21
sonnet-3.7-r-c-0	24.19±.41	76.10±.08	52.85±.19	83.46±.07	41.53±.24
gem-2.0-f-c-2	20.78±.33	76.59±.10	51.38±.20	83.46±.05	36.85±.21
gem-2.0-f-c-5	22.28±.42	73.61±.14	50.18±.18	82.19±.06	30.94±.22
4o-c-4	15.63±.30	78.67±.07	53.50±.18	83.98±.05	36.04±.25
4o-c-5	18.97±.29	77.65±.10	53.44±.20	83.95±.06	36.47±.22
4o-c-6	15.75±.30	78.44±.10	48.53±.16	84.83±.07	33.79±.21
o3-mini-c-0	**.84±.12****	**82.74±.08**	**60.97±.28**	**86.93±.06****	54.22±.25
nfsp	18.35±.46	78.58±.10	55.71±.22	83.76±.07	**55.73±.36**
R-NaD	3.21±.25**	80.56±.09	55.82±.19	85.94±.09**	52.64±.31
aspire	21.00±.37	81.24±.09	58.54±.19	85.74±.06	44.10±.20
soft	29.28±.39	69.35±.10	44.92±.18	78.67±.04	27.31±.17
tough	49.52±.66	61.73±.13	6.49±.05	72.64±.04	.70±.02
walk	51.85±.36	67.97±.01	0.00±.00	75.03±.00	N/A

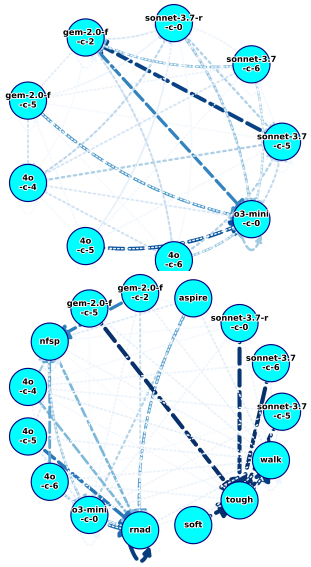


Figure 1: Left: Bootstrap means (± standard errors) for regret, normalized Nash and utilitarian welfare, and EF1 frequency (last three as percentages) in our `LLM-only` and `LLM+baselines` analyses on BGS0; best values are bold. Asterisks on `o3-mini-c-0` and the next-ranked agent mark paired bootstrap significance: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Right: corresponding empirical best-response graphs (top: `LLM-only`; bottom: `LLM+baselines`). EF1 frequency is undefined for `walk` since its negotiations never end in ACCEPT.

`R-NaD`'s significant presence in MENE (see Appendix G), and may also reflect a tendency for algorithms trained in self-play tend to overfit themselves [Hu et al., 2020, Li and Wellman, 2024]. In contrast, `nfsp` does not have a significant self-edge, which is perhaps due to its use of a uniform distribution for opponent modeling [Li and Wellman, 2024]. The probes `soft`, `tough`, and `walk` are weak overall but reveal specific LLM failure modes, discussed below (Section 6.4).

6.3 Behavioral Analysis of Strategies

In BGS0 `LLM-only`, `o3-mini-c-0` performs the best across-the-board; in `LLM+baselines` it remains best on regret (with `R-NaD` a close second) and welfare (`aspire` close behind), and is near the top in EF1.

To understand why, we examine four behavioral metrics:

- *Rounds to termination*: number of offer/response exchanges before a game ends in ACCEPT or WALK.
- *Utility given*: value of items offered to the opponent, measured in the *proposer's* own valuation (i.e., what the proposer gives up).
- *Own utility*.

BGS No Discount (just the BG with no discount is included in these results)					
Agent	Reg↓	NW↑	NW^{+}↑	UW↑	EF1↑
walk	71.36±.30	68.39±.00	0.00±.00	75.40±.00	0.00±.00
tough	55.94±.29	69.49±.01	5.11±.03	77.38±.01	1.99±.04
soft	173.21±.78	68.84±.13	45.19±.26	81.51±.06	26.21±.27
ef1-barg	19.47±.10	83.90±.06	56.88±.20	89.50±.05	**96.16±.22*****
aspire	6.14±.11	86.91±.09	66.63±.28	91.18±.07	61.60±.14
5.2-low	2.40±.14	**90.39±.07**	72.83±.24	**94.38±.06**	60.23±.22
5.2-none	8.10±.23	88.29±.10	66.59±.29	93.01±.08	57.41±.21
5.2-med	6.80±.22	89.53±.08	70.37±.20	93.71±.06	61.45±.21
5.4-low	10.44±.23	88.93±.09	69.43±.24	93.09±.07	57.27±.27
5.4-med	1.43±.08	90.22±.08	**73.31±.23**	94.30±.07	60.09±.23
nfsp	15.96±.08	83.88±.06	55.08±.18	89.35±.05	60.68±.25
ppo	1.94±.08	86.90±.06	63.83±.18	91.66±.05	53.25±.15
psro	**.54±.05**	85.89±.05	60.13±.14	91.19±.04	50.28±.12
mappo	3.28±.10	85.57±.05	63.01±.18	90.66±.04	49.82±.13

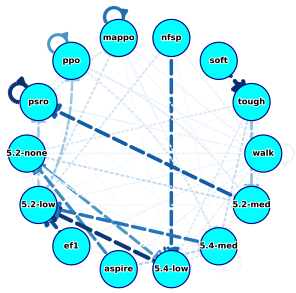


Figure 2: Each cell reports the bootstrap mean ± SE over 1,000 resamples of the empirical game, with metrics evaluated at the MENE equilibrium. Bold marks the best agent per metric. For each metric, we conduct a single paired bootstrap test comparing the best agent to the next-best: both are evaluated on the same resampled game in each bootstrap sample, and the p-value is the fraction of samples in which the next-best agent's score exceeds the best agent's (one-sided, nonparametric, no distributional assumptions). Since exactly one test is conducted per metric, no multiple-testing correction is applied. $*, p < 0.10$, $**, p < 0.05$, $***, p < 0.01$.

- *Concession size*: change in utility given between an agent's first and last offer in multi-offer instances.

In both BGS and BGL, `o3-mini-c-0` ends games fastest with the lowest rounds to termination. In games ending in, ACCEPT it goes 1.40 rounds on average versus the population mean of 1.82; in games ending in WALK, 1.72 versus 2.16. It is also the most generous agent by utility given: 195 on average in BGS and 318 in BGL, compared to population averages of 167 and 285 respectively. It makes attractive offers quickly and either closes or walks early, leaving little room for delay-based exploitation. In `LLM+baselines`, `R-NaD` has the second-lowest regret, near-top welfare, EF1 frequency just below `nfsp`, and is the only identified strong best response to `o3-mini-c-0`. Behaviorally, `R-NaD` bargains more tightly: its average utility given is 169, yet `o3-mini-c-0` narrowly leads in own utility (357 vs. 356 for `R-NaD`), both well above the population mean of 331. Its concession size averages roughly 7 per multi-round game, above the population mean of 4.

6.4 Vulnerability to `tough` and Hallucination

The introduction of `tough` reveals latent vulnerabilities: `tough` is an effective best response to Anthropic Sonnet models in `LLM+baselines`, even though it ranks poorly in MENE regret. In several of the specific game settings (Appendix H) it is also a best response to some Google Gemini models. This indicates that `tough` exploits an excessive credulity present in certain models.

Detailed analysis (see Appendix O) reveals that Anthropic-model agents are far more likely to accept a sub-BATNA offer from `tough` than are other LLM agents.

For `sonnet-3.7-c-5` and `sonnet-3.7-c-6`, roughly half of all turns against `tough` end in acceptance of a sub-BATNA offer. For `sonnet-3.7-r-c-0`, roughly

a quarter do. By contrast, other models terminate turns this way at significantly lower frequencies.

The following example illustrates a typical failure mode for `sonnet-3.7-r-c-0`. The model correctly identifies the incoming offer $[0, 0, 1, 0, 0]$ but hallucinates its own resulting holdings as $[6, 5, 1, 4, 1]$, which inflates the computed value and leads to acceptance of a dominated offer.

Claude Sonnet 3.7 Reasoning Hallucination vs. `tough`

`I must decide whether to accept Player 1's offer, counter, or walk away. Since this is the final round, any counteroffer is treated as WALK.`

Offer from Player 1: `[0, 0, 1, 0, 0]` **My resulting holdings:** `[6, 5, 1, 4, 1]`

Value computation: $(6\times8)+(5\times38)+(1\times7)+(4\times53)+(1\times38) = 495$ Discounted with $0.98^2 \approx 0.9604$: $495 \times 0.9604 = 475.40$

`My outside option value is 201, so accepting is clearly better.`
`"action": "ACCEPT"`

6.5 Provider Level Analysis

Finally, we aggregate from individual models to the provider level and compare Anthropic, Gemini, and OpenAI as families of bargaining agents. Across BG1 through BG6, we select for each provider the model and circle pairing that performs best against an equilibrium mixture of opponent agents from the other two providers, and construct a provider-level meta-game (details in Appendix Q.1). Figure 3 presents the provider-level BR graph.

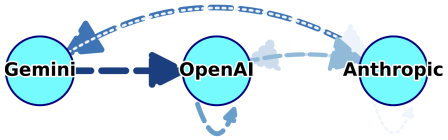


Figure 3: Provider-level empirical best-response graph over BG1–BG6 : Gemini is most often the best response to Anthropic, OpenAI to Gemini, and OpenAI to itself.

The patterns we find align with equilibrium behavior. Anthropic offers the most utility to opponents (in its own valuation), followed by OpenAI, then Gemini (Appendix Q). The overall evidence supports the following account: Anthropic is generous yet more prone to accepting dominated offers, particularly against `tough`. Gemini adopts a tougher posture and exploits this, so it is frequently the best response to Anthropic. OpenAI is more robust to hardline bargaining while remaining reasonably generous, which makes it a strong best response to Gemini and often to itself. In aggregate, this produces provider-specific patterns with a non-transitive cycle of strengths and weaknesses.

7 Conclusion

We performed an in-depth assessment of LLM bargaining as a case study in evaluating the strategic capabilities of LLM agents. A key finding is that competition and cooperation were basically aligned in this mixed-motive setting. OpenAI models, especially `o3-mini`, performed best on both individual and social criteria, and were robust to inclusion of non-LLM baseline strategies. Gemini and especially Claude Sonnet were vulnerable to the hardline `tough` strategy, suggesting that excessive accommodation can be exploited by rigid opponents. Provider-level aggregation surfaced additional, non-transitive patterns among LLM families.

Given the speed of LLM advancement, findings about specific models have a short shelf-life. The more important results from this study are the methodological concepts developed and exercised in the process. By revealing qualitative strategic patterns and unsuspected vulnerabilities, we demonstrated that empirical meta-game analysis provides a practical lens for evaluating multiagent LLM behavior.

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APPENDICES

A Formulation of EF1 condition for additive valuations (Section 3.4)

We begin by recalling the definition of the envy-freeness up to one item (EF1) property for an arbitrary indivisible goods allocation problem. For positive integers m and n, let $\Omega = \{o_1, o_2, \ldots, o_m\}$ be a set of m indivisible items, and $N = \{1, 2, \ldots, n\}$ a set of n agents such that each agent $i \in N$ has a valuation function $v_i : 2^{\Omega} \to \mathbb{R}_{\geq 0}$ satisfying the following conditions:

- $v_i(\emptyset) = 0$;
- $v_i(\Omega) > 0$;
- Monotonicity: $v_i(T) \geq v_i(S)$ whenever $S \subset T \subset \Omega$.

An allocation A is represented by a partition $\{A_1, A_2, \ldots, A_n\}$ of Ω, i.e., $\cup_{i \in N} A_i = \Omega$ and $A_i \cap A_j = \emptyset$ whenever $i \neq j$, such that A_i represents the bundle of items received by agent $i \in N$.

An ordered pair of agents (i, j) is said to satisfy the EF1 condition under allocation A if there exists $S \subseteq A_j$ with $|S| \leq 1$ such that $v_i(A_i) \geq v_i(A_j \setminus S)$. An allocation A is defined as EF1 if every ordered pair of agents satisfies the EF1 condition under A.

The valuation function of an agent i is said to be additive if, for every non-empty $S \subseteq \Omega$, $v_i(S) = \sum_{o \in S} v_{i,o}$ where $v_{i,o} = v_i(\{o\})$ for every $o \in \Omega$. For monotonicity, $v_{i,o} \geq 0$ for every $o \in \Omega$, with the inequality holding strictly for at least one o for each $i \in N$.

Now suppose, under additive valuations, the items are partitioned into T valuation-based equivalence classes called *types* $1, 2, \ldots, T$ and represented by subsets $\{\Omega\}_{t=1}^{T}$ such that:

- $\Omega_t \neq \emptyset$ for each $t \in \{1, 2, \ldots, T\}$;
- $\cup_{t=1}^{T} \Omega_t = \Omega$;
- $\Omega_t \cap \Omega_{t'} = \emptyset$ whenever $t \neq t'$;
- for each agent $i \in N$ and each type $t \in \{1, 2, \ldots, T\}$, $v_{i,o} = v_{i,o'}$ whenever $o, o' \in \Omega_t$.

In this scenario, we can equivalently represent the item set Ω in terms of the quantity vector $q = (q_1, q_2, \ldots, q_T)$, where $q_t = |\Omega_t|$ for every type t; we can also express agent i's valuation as a T-vector $v_i = (v_{i,1}, v_{i,2}, \ldots, v_{i,T})$ where, for every t, $v_{i,t} \coloneqq v_{i,o}$ for each $o \in \Omega_t$; an allocation A can be represented as a collection of T-vectors $(a_1, a_2, \ldots, a_n)$ such that $a_{i,t}$ is the number of items of type t received by agent i under allocation A. Clearly, $a_{i,t}$ is a non-negative integer for each i, t with $\sum_{i \in N} a_i = q$.

Under this representation, agent i's additive valuation under allocation A can be alternatively expressed as $v_i(A_i) = \sum_{t=1}^{T} v_{i,t} a_{i,t} = v_i \cdot a_i$, the scalar product.

Proposition A.1. *In a scenario with additive valuations and item types as described above, an ordered pair of agents (i, j) satisfies the EF1 condition under an allocation A where $a_{j,t} > 0$ for at least one t if and only if*

$$v_i \cdot a_j - v_i \cdot a_i \leq \max_{t:a_{j,t}>0} v_{i,t}.$$

Proof. First note that $\max_{t:a_{j,t}>0} v_{i,t}$ is well-defined and non-negative since $a_{j,t} > 0$ for at least one t.

We will first prove the forward direction: the general EF1 condition implies the inequality in the theorem statement. From the general definition of the EF1 condition, there are two cases:

Case I. $\exists S.|S| = 0 \wedge v_i(A_i) \geq v_i(A_j \setminus S)$.

This implies that

$$\begin{aligned} v_i(A_i) &\geq v_i(A_j \setminus \emptyset) = v_i(A_j) \\ \text{i.e., } v_i(A_j) - v_i(A_i) &\leq 0 \\ \text{i.e., } v_i \cdot a_j - v_i \cdot a_i &\leq 0 \leq \max_{t:a_{j,t}>0} v_{i,t} \end{aligned}$$

In fact, in this case, agent i is envy-free of agent j.

Case II. $\exists S.|S| = 1 \wedge v_i(A_i) \geq v_i(A_j \setminus S)$.

This means that there is an item $o' \in A_j$ such that

$$v_i(A_i) \geq v_i(A_j \setminus \{o'\}) = \sum_{o \in A_j \setminus \{o'\}} v_{i,o} = \sum_{o \in A_j} v_{i,o} - v_{i,o'} = v_i(A_j) - v_{i,o'},$$

$$\text{i.e., } v_i(A_j) - v_i(A_i) \leq v_{i,o'}.$$

By definition, $v_{i,o'} \leq \max_{o \in A_j} v_{i,o}$.

It is also evident that $\max_{o \in A_j} v_{i,o} = \max_{t:a_{j,t}>0} v_{i,t}$ since they are two ways of expressing agent i's maximum value for any item in agent j's bundle. This completes the proof of the forward direction.

For the reverse direction, pick a type $t^* \in \arg\max_{t:a_{j,t}>0} v_{i,t}$ and then an item o^* of type t^* in A_j. We know that such an item exists since $a_{j,t^*} > 0$. Clearly, $v_{i,o^*} = \max_{t:a_{j,t}>0} v_{i,t}$. Thus, the inequality in the theorem statement implies that

$$\begin{aligned} v_i(A_j) - v_i(A_i) &\leq v_{i,o^*} \\ \text{i.e., } v_i(A_i) \geq v_i(A_j) - v_{i,o^*} &= \sum_{o \in A_j} v_{i,o} - v_{i,o^*} = \sum_{o \in A_j \setminus \{o^*\}} v_{i,o} = v_i(A_j \setminus \{o^*\}), \end{aligned}$$

which gives us a subset $S = \{o^*\}$ for which the generic definition of EF1 is satisfied. □

Note that, if $\forall t.a_{j,t} = 0$, then $A_j = \emptyset$ so that $v_i(A_j) = 0 \leq v_i(A_i)$. Hence, (i, j) trivially satisfy the EF1 condition; in fact, i is envy-free of j.

B Experimental setup details

Game class	T	Game	γ	R	#settings in cohort	Total #settings
BGL	5	BG1	0.9	3	$11,630$	$27,413$
		BG2	0.98	3	$7,933$	
		BG3	0.98	5	$7,850$	
BGS	3	BG4	0.9	3	$25,700$	$75,740$
		BG5	0.98	3	$24,900$	
		BG6	0.98	5	$25,100$	

Table 1: Statistics of game instances generated for our bootstrapped empirical meta-game analysis, as described in Section 5. T denotes the number of item types in the pool. q is the T-length vector of the numbers of items of all types in the pool: for a BGL game instance, we sample $q_t \sim \texttt{Poisson}(4)$ for each $t \in T$; for BGS, we fix $q = (7, 4, 1)$. γ is the (exponential) discount rate shared by both players. R is the number of bargaining rounds in an instance.

C Negotiation mistakes

In any bargaining interaction, certain moves should be avoided to maintain rational play. We identified five key mistakes that we describe below. For the illustrative examples, assume an instance with two item types for which your private values are given by the vector $(4, 2)$, and each offer is represented by a vector of the numbers of items of the two types you will receive if the offer is accepted, regardless of who makes the offer.

1. **M1: Making an offer less valuable to you than the latest offer you received.**

   Example: Your opponent offers you $(3, 0)$, which is worth $3 \cdot 4 + 0 \cdot 2 = 12$ to you. You then counter with $(2, 1)$, which is worth $2 \cdot 4 + 1 \cdot 2 = 10$ to you, strictly less than what you could have realized by accepting the opponent's offer instead.

2. **M2: Making an offer less valuable to you than your BATNA.** This is an unreasonable move because you know that you could have realized a better value if you had walked away, letting negotiations fail, than if your opponent accepts your offer under consideration.

   Example: Your BATNA is 10, and your offer is $(1, 1)$, worth $1 \cdot 4 + 1 \cdot 2 = 6 < 10$.

3. **M3: Offering no items or all items.** An offer where you keep all items for yourself is strategically imprudent as it is unlikely to be accepted by the opponent and also patently uncooperative in our games where it is generally possible to realize a value above one's BATNA without grabbing all items for yourself. The other extreme of yielding all items is also ill-advised because, if accepted, it ensures a zero value for you (below your BATNA).

4. **M4: Accepting an offer less valuable to you than your BATNA.** The argument is similar to that for **M2**.

   Example: Your BATNA is 10, but you accept when your opponent offers $(1, 1)$, which is worth $1 \cdot 4 + 1 \cdot 2 = 6 < 10$ to you.

5. **M5: Walking away from an offer that is more valuable to you than your BATNA.** This action is strictly dominated by accepting the offer under consideration.

   Example: Your BATNA is 9. The opponent offers you $(2, 1)$, worth $2 \cdot 4 + 1 \cdot 2 = 10$. Walking away in response leaves you with $9 < 10$.

D Prompt engineering for negotiation

This section describes each incremental "circle" of prompts provided to the negotiation models. Each circle adds guidance or strategic considerations aimed at progressively improving the reasoning and negotiation outcomes. The circles follow a progression, with each level introducing more sophisticated negotiation concepts.

The negotiation history is maintained throughout the game, displaying each player's actions chronologically. The current offer on the table is explicitly stated, and the action prompt varies based on whether the agent is Player 1 making the first move, has a current offer to consider, or needs to make a counteroffer.

Prompts for the 3 item games are modified in accordance with the differing item count in its games.

D.1 System Prompt

All LLM agents receive a consistent system prompt that establishes their role as negotiators and enforces the required response format:

System Prompt

```
You are an AI negotiator participating in a negotiation
game.
You must respond in one of these formats only:
  1. {"action": "ACCEPT"}
  2. {"action": "WALK"}
  3. {"action": "COUNTEROFFER", "offer": [n1, n2, ...]}
Ensure your response is valid JSON and matches one of
these exact formats.
```

This system prompt is sent with every API call across all models and prompt circles, ensuring consistent response formatting. The detailed game instructions and strategic guidance are provided in the user message, which varies according to the prompt circle level described below.

D.2 Prompting "Circles"

Circle 0: The Vestibule (Setting the Stage). The baseline negotiation scenario provides the foundational rules of the negotiation game, including player identities, item quantities, private valuations, outside offers, and negotiation structure. No additional guidance is provided beyond the basic game mechanics.

Circle 0 Prompt

```
You and another agent have to negotiate a division
of items between the two of you.  You are Player
{my_player_num} and the other agent is Player
{other_player_num}.  There are T types of items, called
item 1 through item T. There are {quantities} to divide.
Both you and Player {other_player_num} have a private
value per unit of each item type.  These values are drawn
from a uniform random distribution, ranging from 1 to V-1.
Your private values are {values}.  You have a private
outside offer drawn from a uniform random distribution
ranging from {p1_outside_offer[0]} to your total value
of all items, which is {p1_outside_offer[1]}.  Player
{other_player_num} has a private outside offer drawn from
a uniform random distribution ranging from 1 to their
total value of all items.  Your outside offer value is
{w}.
The negotiation proceeds in {R} rounds.  There is a
discount rate gamma = {g}, such that if the process
concludes after r rounds the overall value of the
negotiation to each player is their value for the outcome
multiplied by gamma to the power (r-1).  At each round,
Player 1 takes an action, followed by Player 2.  The
possible actions are to ACCEPT the other player's current
offer (if any), make a COUNTEROFFER, or WALK away.  If the
game gets to the last round, and player 2 chooses to make
a counteroffer, this is treated as a WALK. If a player
chooses ACCEPT, the negotiation ends in a deal to divide
the items according to the accepted offer.  The value of
an outcome is determined by each player's private values
per unit of each item and the quantities they receive
in the deal.  This value is adjusted by the discount
factor, which is used to compute the present value of
the negotiation outcome.  If a player chooses WALK, the
negotiation ends without a deal, and each player receives
the value of their private outside offer.
```

Circle 1: The Limbo of Objectives. Circle 1 explicitly introduces the negotiation objective, reminding the agent to maximize its negotiation outcome and to carefully consider its guaranteed alternative, the outside offer.

Circle 1 Prompt

```
[All instructions from Circle 0, plus:]
Your objective is to maximize your value of the outcome
of the negotiation game.  Remember, you have a guaranteed
alternative:  your outside offer.
```

Circle 2: The Circle of Calculations. Circle 2 provides concrete numerical examples illustrating how agents should evaluate potential counteroffers by explicitly calculating their total value, ensuring the counteroffers are superior to their outside offers.

Circle 2 Prompt

```
[All instructions from Circle 1, plus:]
Before making any counteroffer, you should calculate its
total value to you and compare it to your outside offer
value of w.  For example, if you were considering offering
the other player 2 units of each item (keeping 3 units of
each for yourself), you would calculate:  3 units of item
1 = 3 × values[0] = 3 × values[0] (multiplying units by
your value per unit) 3 units of item 2 = 3 × values[1]
= 3 × values[1] (multiplying units by your value per
unit) 3 units of item 3 = 3 × values[2] = 3 × values[2]
(multiplying units by your value per unit) 3 units of item
4 = 3 × values[3] = 3 × values[3] (multiplying units by
your value per unit) 3 units of item 5 = 3 × values[4] = 3
× values[4] (multiplying units by your value per unit)
Total value = sum([3 × values[i] for i in range(T)]) (sum
of all item values)
```

Circle 3: The Circle of Systematic Analysis. Circle 3 introduces a systematic three-step questioning strategy designed to facilitate a comprehensive analysis:

1. Analysis of the current negotiation situation
2. Assessment of the value of current and potential offers
3. Decision-making based on comparative analyses

Circle 3 Prompt

```
[All instructions from Circle 2, plus:]
The following step-by-step questions are designed to guide
you through a comprehensive analysis.  By systematically
addressing these questions, you can evaluate the current
state of the negotiation, assess potential offers, and
make informed decisions.  You must use the information
that you acquired through the step-by-step questioning
above to decide what action you will make.
Let's walk through this step by step:

  1. First, analyze the current situation:

       • What is my outside offer value?
       • What are the values of the items involved?
       • What is the total pool of items?
       • How does the discount factor influence the value
         of accepting the current offer versus waiting for
         future offers?

  2. Assess the value of offers:

       • For the current offer (if any):  What is my total
         value if I accept it?
       • For potential counteroffers:  What would be my
         total value for different proposed divisions?
       • How do these values compare to my outside offer
         value?

  3. Make a decision based on the analysis:

       • Should I accept the current offer?
       • Should I walk away and take my outside offer?
       • Or should I propose a specific counteroffer?
```

Circle 4: The Circle of Common Errors. Circle 4 identifies five common negotiation mistakes that conflict with the negotiation objectives:

- **Mistake 1:** Making an offer worse than your previous offer. This occurs when you reject an offer better for you than the one you subsequently propose.
- **Mistake 2:** Making an offer worse for you than your outside offer. This happens if you propose giving away so much that what you keep is worth less than your guaranteed alternative.
- **Mistake 3:** Offering no items or all items. Offering nothing (or everything) to the opponent (in the early or middle rounds) can be a clear suboptimal move.

- **Mistake 4:** Accepting an offer worse for you than your outside offer. This occurs if you accept a division that yields a payoff lower than your guaranteed fallback.
- **Mistake 5:** Walking away from an offer better than your outside offer. This occurs when you reject a division that actually yields a higher payoff than your fallback.

Circle 4 Prompt

```
[All instructions from Circle 3, plus:]
In the bargaining game, there are five mistakes you
can make that conflict with your objectives.  While
these aren't the only possible errors, they represent
undesirable negotiation behaviors that can undermine your
payoff or cause you to miss out on better deals.

  1. Mistake 1:  Making an offer worse than your previous
     offer.  This occurs when you reject an offer better
     for you than the one you subsequently propose.
  2. Mistake 2:  Making an offer worse for you than your
     outside offer.  This happens if you propose giving
     away so much that what you keep is worth less than
     your guaranteed alternative, which is your outside
     offer.
  3. Mistake 3:  Offering no items or all items.  Offering
     nothing (or everything) to the opponent (in the early
     or middle rounds) can be a clear sub-optimal move.
  4. Mistake 4:  Accepting an offer worse for you than
     your outside offer.  This occurs if you accept
     a division that yields a payoff lower than your
     guaranteed fallback.
  5. Mistake 5:  Walking away from an offer better than
     your outside offer.  This occurs when you reject a
     division that actually yields a higher payoff than
     your fallback.
```

Circle 5: The Circle of Error Prevention. Circle 5 provides explicit examples and strategic recommendations to prevent the identified mistakes, emphasizing the calculation of offer values and iterating proposals until surpassing the outside offer value. The specific example is dynamically generated for each player in each game instance. This circle introduces a concrete example of a suboptimal offer to illustrate the importance of value calculation.

Circle 5 Prompt

```
[All instructions from Circle 4, plus:]
To prevent these mistakes, adopt a strategy similar to
the following example:  before making any counter-offer,
calculate its total value to you and compare it to your
outside-offer value.  For instance, suppose you keep only
{example_offer_less_than_outside_offer_self} items and
offer the rest to the other party.  Your value would be:


{values[0]} × {example_offer_less_than_outside_offer_self[0]} +
{values[1]} × {example_offer_less_than_outside_offer_self[1]} +
{values[2]} × {example_offer_less_than_outside_offer_self[2]} +
{values[3]} × {example_offer_less_than_outside_offer_self[3]} +
{values[4]} × {example_offer_less_than_outside_offer_self[4]}

which is {np.dot(values, example_offer_less_than_outside_offer_self)}
(sum of all item values) and is less than your outside
offer of {w}.  If your proposed offer results in a value
lower than your outside offer, continue iterating until
you develop a more advantageous offer that is better than
your outside offer.
This reasoning can be applied to each of the five
highlighted mistakes to ensure that your offers align
with your objectives and avoid undesirable negotiation
behaviors.
```

Circle 6: The Circle of Strategic Inference. Circle 6 introduces strategic inference about the opposing agent's valuations based on their negotiation behavior. Agents are guided to infer relative valuations from the distribution of items in the opponent's offers, leveraging this information to inform their own negotiation strategy.

Circle 6 Prompt

```
[All instructions from Circle 5, plus:]
Keep in mind the offers the opposing agent makes reflects
its own values.  If their offer includes most or all units
of a particular item, it might indicate that the agent
does not highly value that item, whereas offering none
could suggest the opposite.  You can use this kind of
evidence to help inform your decision-making.
```

D.3 Response format and history tracking

All prompts include standardized action specification requirements and maintain negotiation history across rounds:

Response Format

```
Please show your reasoning step by step, then provide your
action in one of these formats in your response (if you do
not do this your response will be invalid):
{"action": "ACCEPT"} -- to accept the current offer
{"action": "WALK"} -- to walk away from negotiations
{"action": "COUNTEROFFER", "offer": [n1, n2, ...]} --
where n1, n2, ... are numbers representing the number of
units of each item being offered to the other player as
part of the counter-offer.
Any response not in these exact formats will be invalid
and treated as a WALK. If you provide a counter-offer, it
must be a valid offer; otherwise it will be treated as a
WALK.
It is now round {r}.
Negotiation history: [History of all previous rounds,
including offers made and actions taken]
Current offer on the table (the amount of each item being
offered to you): {current_offer}
What is your action? [Action prompt specific to the
current game state]
```

E Circle selection methodology

We paired each of the LLMs GPT-4o, o3-mini, Anthropic Claude 3.7 Sonnet, and Gemini 2.0 Flash with each of our 7 prompt circles (as described in Section D) and sampled 100 self-play instances of BGL. For each instance, we recorded the number of times each mistake (as defined in Section C) was committed by both players over all rounds. For each LLM, we use a radar chart to visualize the cumulative frequencies of all 5 mistakes over all 100 instances for each circle in Fig. 4.

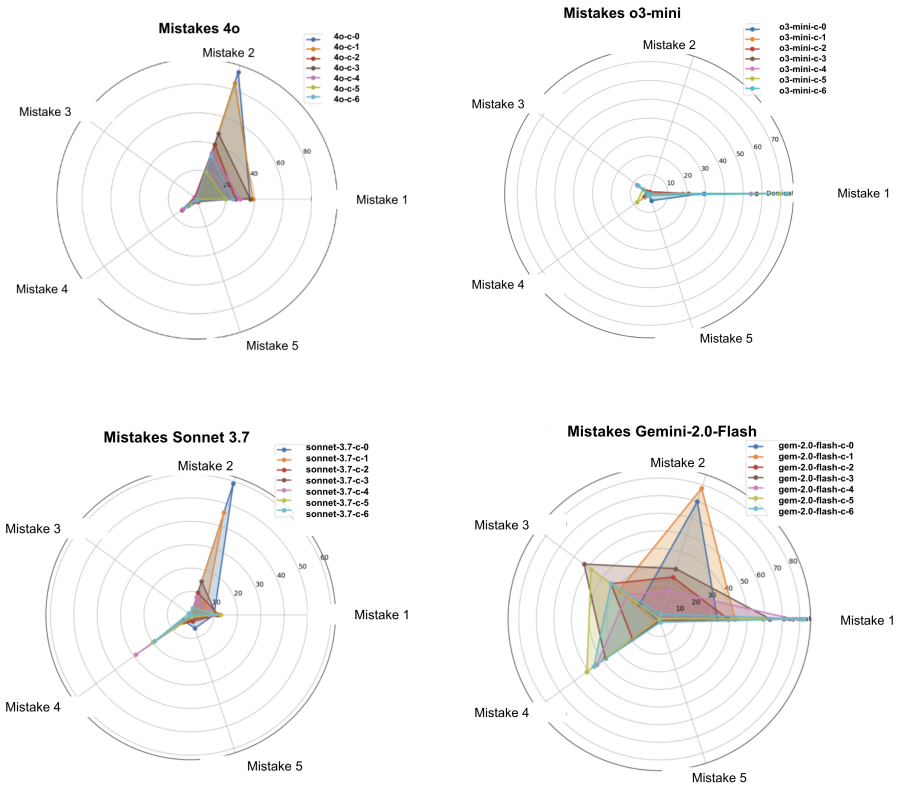


Figure 4: **Mistake counts for each negotiation strategy as a function of prompt "circle."** Each radar chart shows the number of times mistakes of each kind (as defined in Section C) were made by the corresponding LLM when paired with each incremental prompt circle $(0-6)$ over all self-play histories. Lower counts indicate better adherence to rational negotiation behavior.

Although we crafted our prompt circles to provide progressively more elaborate guidance to the LLM to become a better negotiator (in particular, to avoid the potential mistakes we identified), our intuition was not generally borne out by our experimental results, so we finally had to make a judgment call to pick our final set LLM-circle pairings for our meta-game analysis. GPT-4o and Claude Sonnet 3.7 achieved a sub-

stantial reduction in mistake frequencies with higher circles, although not a consistent Pareto improvement over all mistakes with an increase in the circle number. Gemini 2.0 Flash displayed minimal sensitivity to prompt complexity. Most surprisingly, the performance OpenAI's reasoning model o3-mini along most dimensions significantly deteriorated with higher circles, suggesting that additional instruction may interfere with their inherent reasoning capabilities. In light of this last finding, we did not subject Sonnet 3.7 Reasoning to the same treatment as the above 4 models and used its pairing with Circle 0 only.

Our final 9 LLM agents are:

- `4o-c-4`,
- `4o-c-5`,
- `4o-c-6`,
- `o3-mini-c-0`,
- `sonnet-3.7-c-5`,
- `sonnet-3.7-c-6`,
- `sonnet-3.7-r-c-0`,
- `gem-2.0-f-c-2`,
- `gem-2.0-f-c-5`

F Aspiration negotiator

Our heuristic strategy `aspire` (Section 4) closely follows the implementation of the aspiration-based negotiation agent in the NegMAS github repository (https://github.com/yasserfarouk/negmas) Mohammad et al. [2020].

Setup

- Rounds $r \in \{1, \dots, R\}$ with $R \in \{3, 5\}$.
- Discount rate $\gamma \in \{0.9, 0.98\}$.
- $asp_{\max} \in [0, 1]$ is a ceiling fraction. We have it set to 0.85 by default (NegMAS uses 1.0). It never becomes the opening target directly, instead we blend it with the sampled BATNA in every game (details below). This way our opening target scales with the BATNA.
- θ_r is the raw-utility target produced by the concession schedule. Given BATNA B and total value $U_{\max}$ we compute

$$\theta_1 = B + s(\gamma) \cdot asp_{\max} \cdot (U_{\max} - B)$$

  where the scale $s(\gamma)$ is a piecewise constant depending on the discount rate:

$$s(\gamma) = \begin{cases} 0.15 & \text{if } \gamma \leq 0.92 \\ 0.25 & \text{if } 0.92 < \gamma \leq 0.96 \\ 0.40 & \text{if } \gamma > 0.96 \end{cases}$$

  This keeps θ_1 in $[B, U_{\max})$, with patient games (high γ) starting at higher aspiration levels.
- $\epsilon = 0.05$ is the acceptance slack: at round r, we shrink the target by 5%,

$$\theta_r^{\text{eff}} = \max\big(\theta_r - \epsilon,\, B\big),$$

  so we never demand less than the BATNA.

Concession Schedules

After the dynamic opening value is set, later rounds follow a NegMAS curve that slides monotonically back to the BATNA. The exponent p is chosen from NegMAS presets based on γ, coupled with the initial scale $s(\gamma)$:

γ	s	Profile	p (poly)
≤ 0.92	0.15	conceder	0.25
$(0.92, 0.96]$	0.25	linear	1.0
> 0.96	0.40	Boulware	4.0

This coupling ensures aggressive early concession when discounting is steep, and patient holding-out when future payoffs are nearly undiscounted. Let $\theta_R = B$ and $\Delta = \theta_1 - \theta_R$.

Polynomial family

$$\theta_r = \theta_R + \Delta \left(1 - \left(\frac{r-1}{R-1}\right)^p\right), \qquad p \in \{4.0 \text{ (Boulware)}, \ 1.0 \text{ (linear)}, \ 0.25 \text{ (conceder)}\}.$$

Exponential family

$$\theta_r = \theta_R + \Delta \left(\frac{e^{(1-\frac{r-1}{R-1})^p} - 1}{e-1}\right), \qquad p \in \{0.125 \text{ (Boulware)}, \ 0.725 \text{ (linear)}, \ 4.0 \text{ (conceder)}\}.$$

Decision rules

Target selection

- Compute θ_r using the chosen schedule; define $\theta_r^{\text{eff}} = \max(\theta_r - \epsilon, B)$.
- Player 1 "leads" the schedule: in round r it uses $\theta_{r+1}^{\text{eff}}$ (clamped at $r = R$); Player 2 uses θ_r^{eff}.
- Targets remain in raw utility while constructing counter-offers; discounts are applied only when evaluating incoming offers against the target and BATNA.

Accept / Walk

- ACCEPT iff the discounted value of the opponent's offer meets both the discounted target and the discounted BATNA.
- WALK if every legal counter-offer that clears the target would concede zero items to the opponent (such offers are filtered out). Practically, this handles the high-BATNA edge cases where there is no room to bargain/concede.

Counter-offer construction

1. **Mirror the opponent's keep.** If the last opponent offer was x, their keep vector is $k = q - x$ for total quantities q. We search for k_y solving

$$\min_{k_y \in \mathcal{K}} \sum_{j=1}^{T} |k_{y,j} - k_j| \quad \text{s.t. } u_i(k_y) \geq \theta_r^{\text{eff}}, \ 0 \leq k_{y,j} \leq q_j,$$

   where $\mathcal{K}$ is the set of keep vectors induced by legal actions.

2. A depth-first search enumerates counts that satisfy the utility floor while minimizing the ℓ_1 distance, ties go to the candidate closest to the target, and then to higher utility.

3. We're using the opponent's last offer as an opponent model, where we are trying to make them an offer that is as close as possible to their prior offer (in terms of the items they kept) while also adhering to our target.

We instantiate the Aspiration agent using the polynomial family with a maximum aspiration ceiling of 85% of the surplus above the BATNA. Acceptance slack is set to 5%, meaning the agent will accept offers slightly below its target threshold. The initial aspiration scale and concession exponent are coupled to the discount factor: patient games ($\gamma = 0.98$) use a Boulware profile that holds out early and concedes slowly, while impatient games ($\gamma = 0.9$) use a conceder profile that makes aggressive early concessions.

G Empirical Game Average Max-Entropy Nash Equilibria

In this section, we report the average maximum-entropy Nash equilibrium computed on every cohort of bootstrap samples in `LLM-only` and `LLM+baselines`.

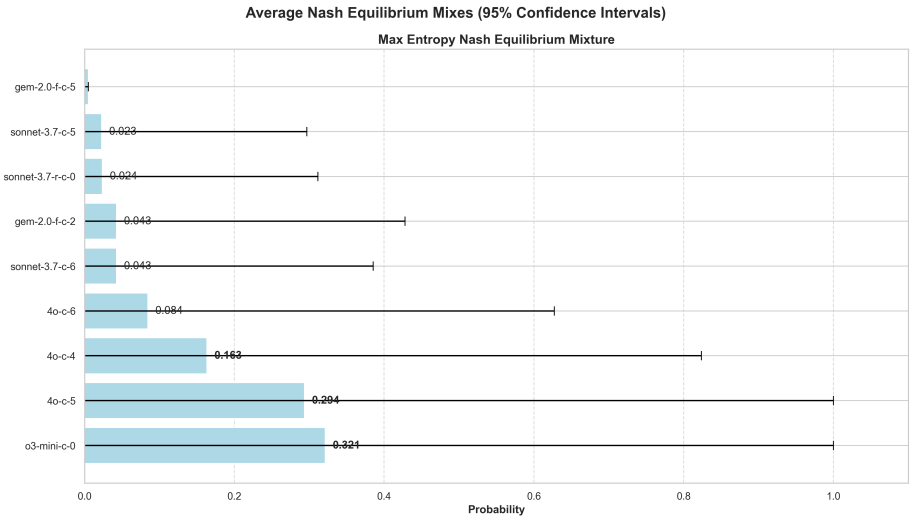


Figure 5: BG1, `LLM-only`

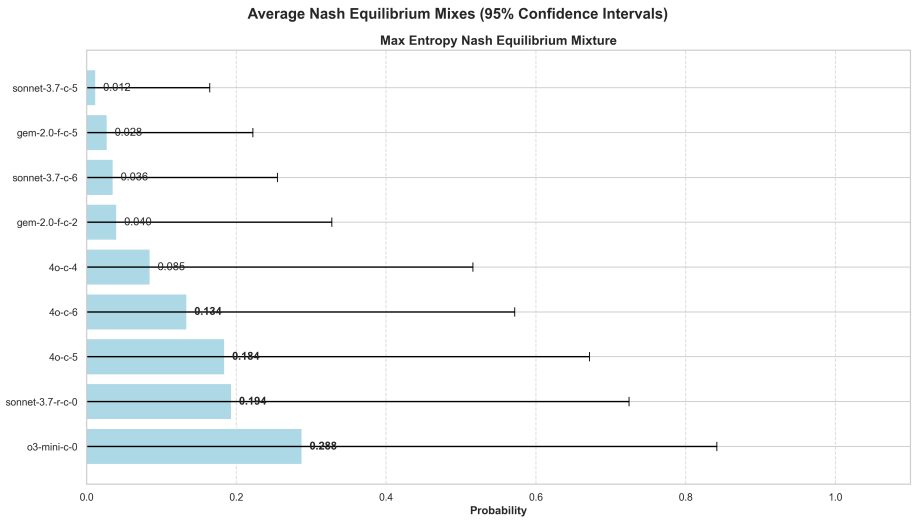


Figure 6: BG2, `LLM-only`

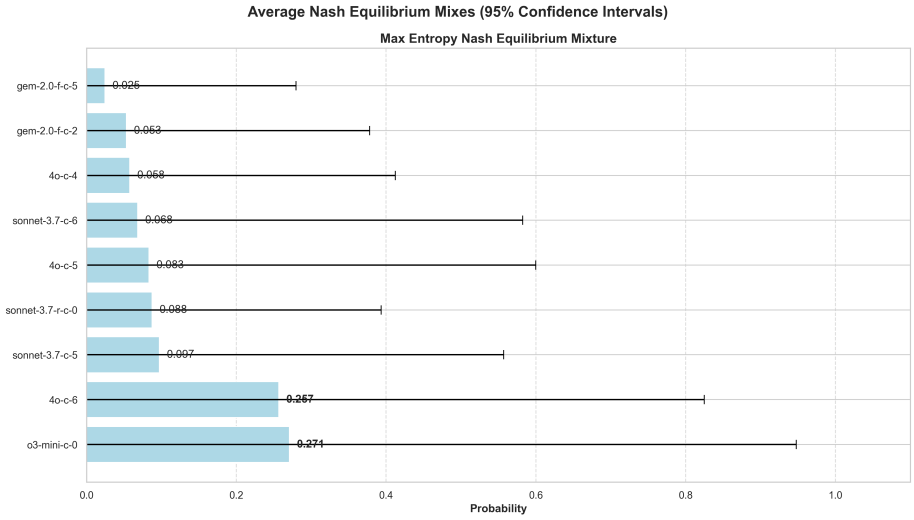


Figure 7: BG3, `LLM-only`

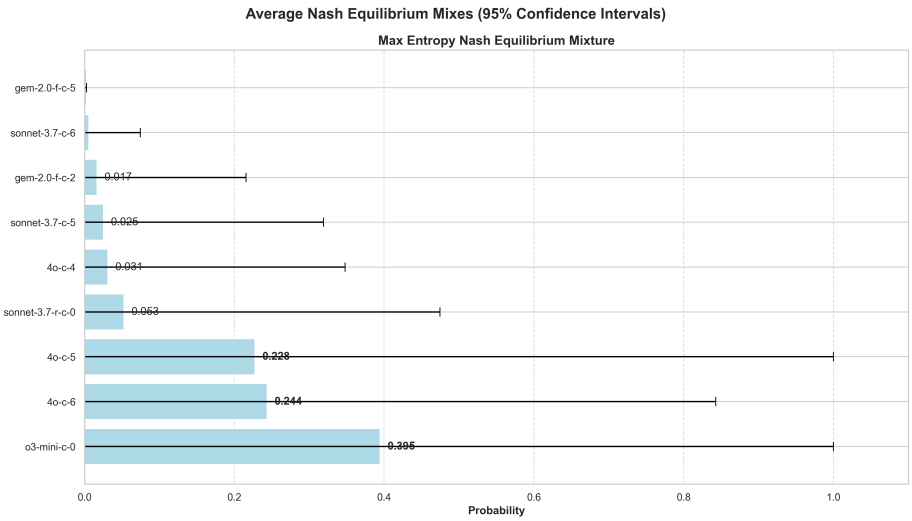


Figure 8: BGL0, `LLM-only`

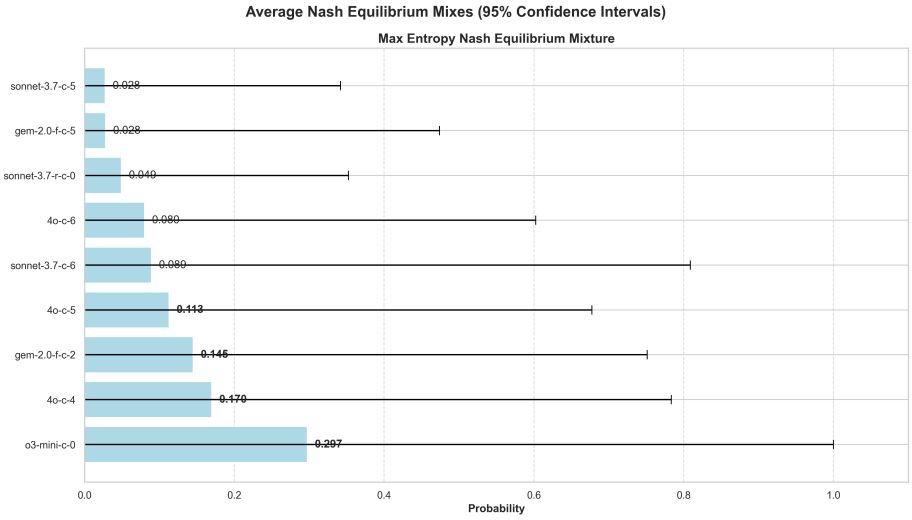


Figure 9: BG4, `LLM-only`

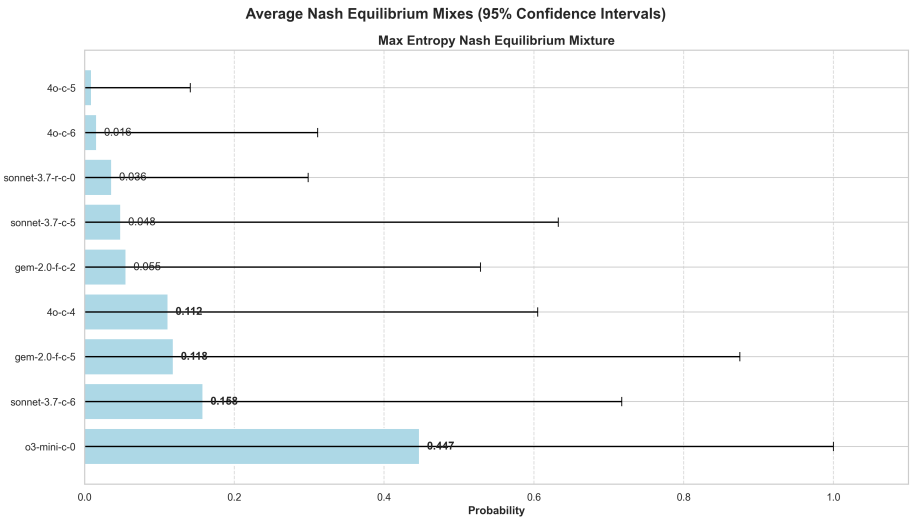


Figure 10: BG5, `LLM-only`

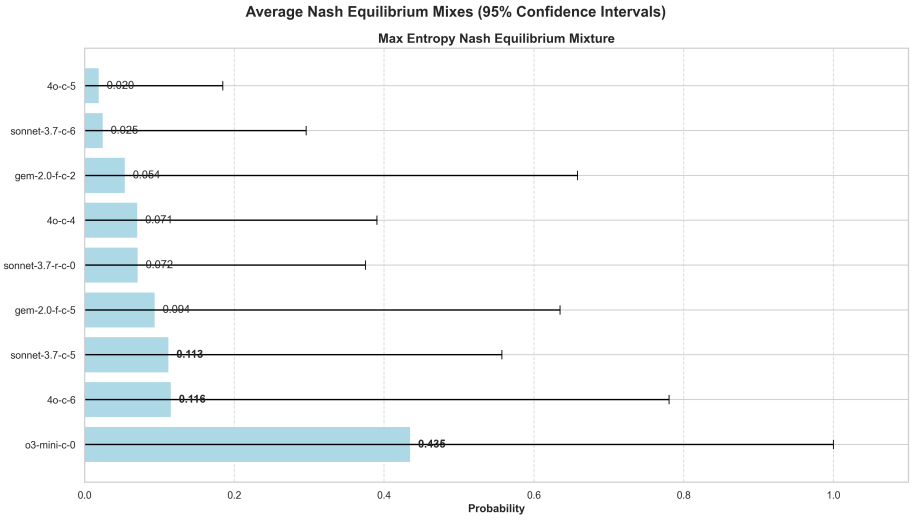


Figure 11: BG6, `LLM-only`

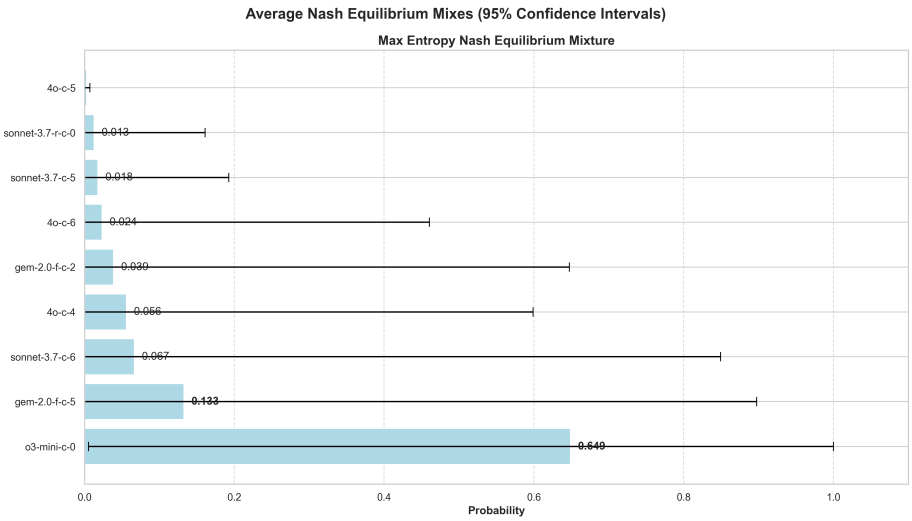


Figure 12: BGS0, `LLM-only`

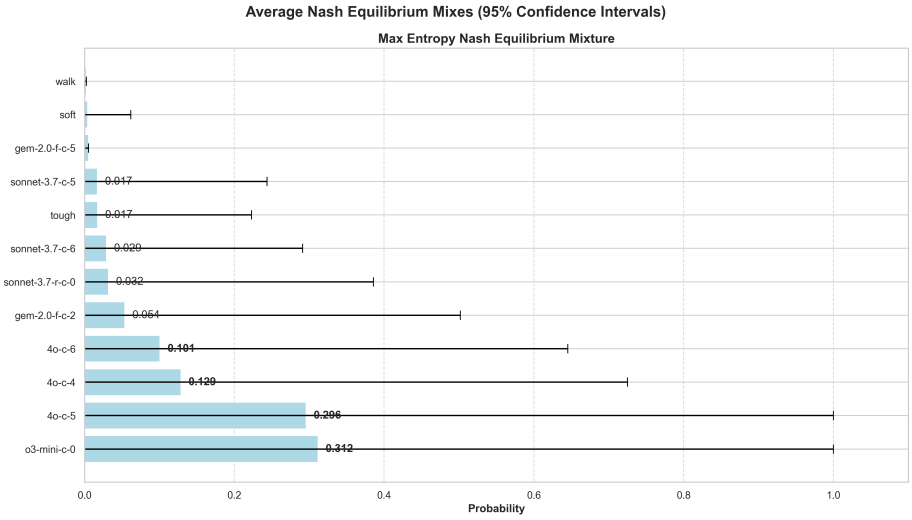


Figure 13: BG1, `LLM+baselines`

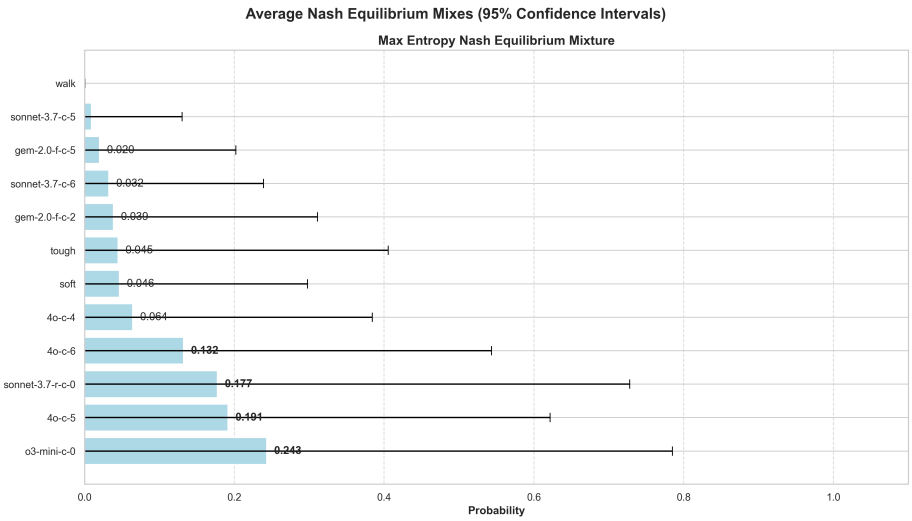


Figure 14: BG2, `LLM+baselines`

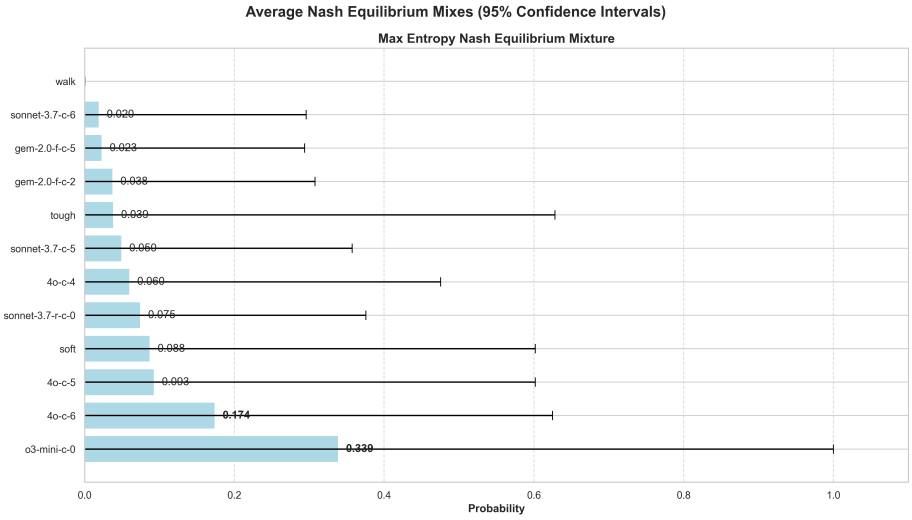


Figure 15: BG3, `LLM+baselines`

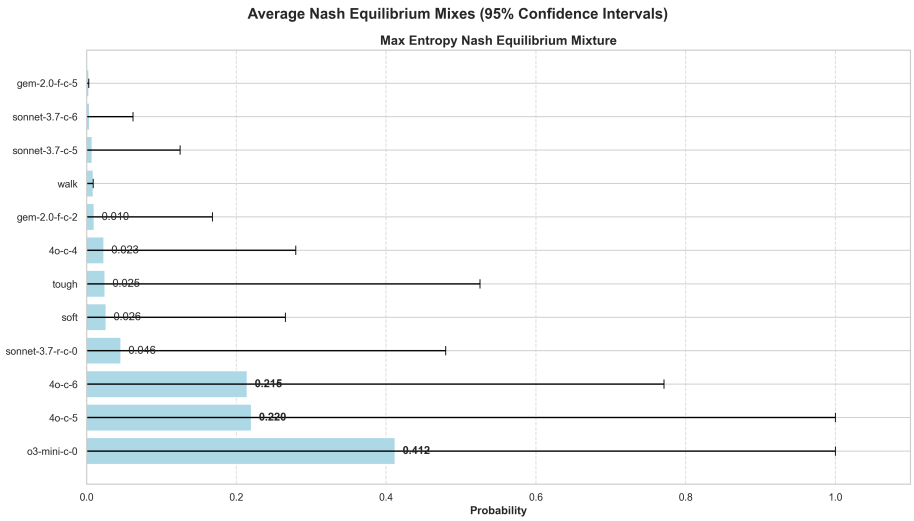


Figure 16: BGL0, `LLM+baselines`

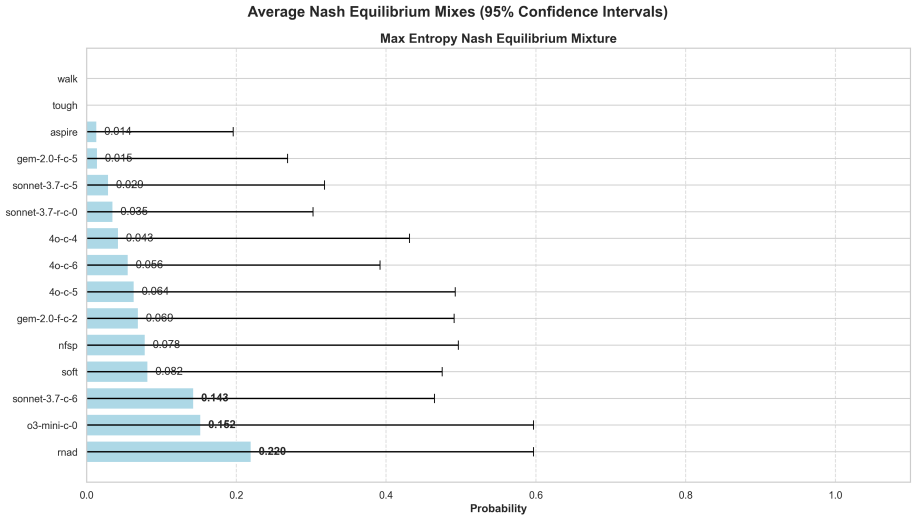


Figure 17: BG4, `LLM+baselines`

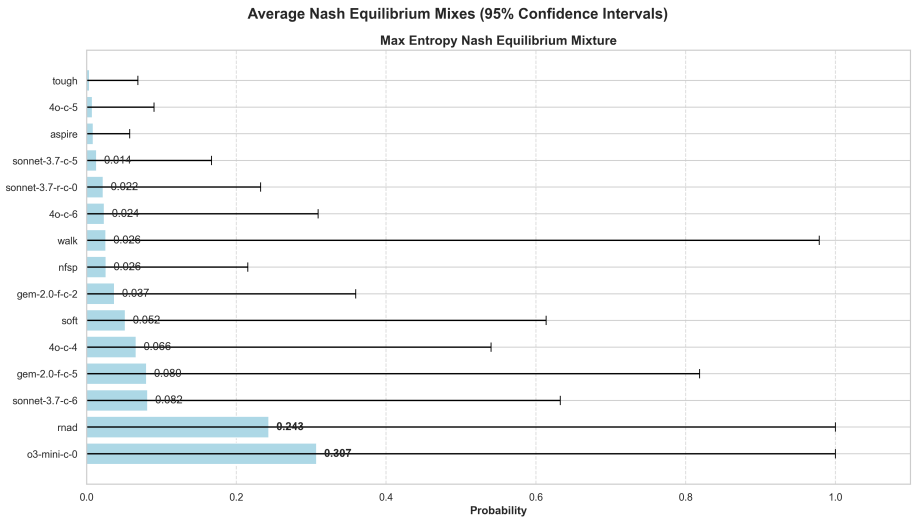


Figure 18: BG5, `LLM+baselines`

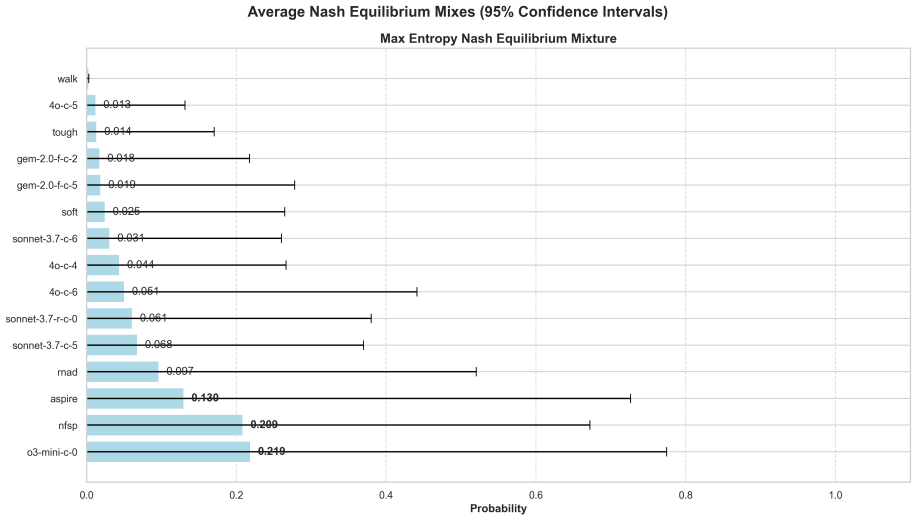


Figure 19: BG6, `LLM+baselines`

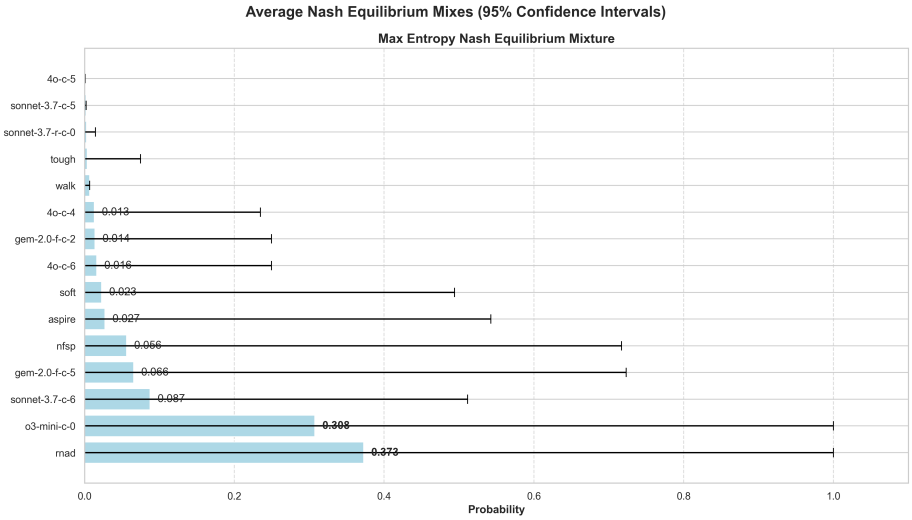


Figure 20: BGS0, `LLM+baselines`

H Disaggregated bootstrapping results for `LLM-only` and `LLM+baselines`

Table 2: Bootstrap means ($\pm$ standard errors) of regret, normalized Nash, NW$^+$, and utilitarian welfares, and EF1 frequencies, the last 4 expressed as percent points, for our `LLM-only` analysis on BG1–3. Lower values are better for regret and worse for the other metrics. The best value of each metric is in boldface.

Agent	BG1 ($\gamma = 0.9, R = 3$)					BG2 ($\gamma = 0.98, R = 3$)				
	MENE Regret	NW%	NW$^+$ %	UW %	EF1 %	MENE Regret	NW%	NW$^+$ %	UW %	EF1 %
sonnet-3.7-c-5	34.95±0.76	89.19±0.08	52.12±0.16	86.89±0.07	33.85±0.17	37.11±0.86	87.67±0.07	55.44±0.14	85.68±0.06	35.35±
sonnet-3.7-c-6	20.77±0.62	90.93±0.08	48.75±0.16	87.79±0.06	33.05±0.19	17.71±0.63	90.93±0.08	58.34±0.18	86.05±0.07	36.83±
sonnet-3.7-r-c-0	39.79±0.88	87.66±0.07	45.98±0.13	84.11±0.05	41.71±0.15	15.93±0.78	89.44±0.08	53.56±0.16	85.52±0.06	44.56±
gem-2.0-f-c-2	38.31±0.89	84.76±0.07	42.40±0.13	82.47±0.06	34.79±0.17	28.71±0.82	85.28±0.08	49.60±0.17	82.96±0.06	33.64±
gem-2.0-f-c-5	48.90±0.71	84.01±0.10	43.53±0.15	84.01±0.06	26.81±0.23	29.01±0.85	83.74±0.10	45.01±0.15	84.71±0.07	33.01±
4o-c-4	13.27±0.56	91.97±0.08	49.14±0.13	86.62±0.06	37.88±0.20	13.27±0.56	90.42±0.08	52.26±0.18	85.72±0.07	41.80±
4o-c-5	6.05±0.40	**93.00 ± 0.08**	50.96±0.18	**88.14 ± 0.06**	37.38±0.21	9.37±0.57	90.46±0.08	55.07±0.17	85.87±0.06	40.54±
4o-c-6	20.49±0.67	92.83±0.07	54.34±0.17	87.66±0.05	38.86±0.26	18.49±0.74	89.33±0.06	59.28±0.18	85.38±0.05	38.38±
o3-mini-c-0	**4.68 ± 0.35**	92.48±0.08	**56.70±.20**	86.73±0.06	**52.20 ± 0.19**	**2.73 ± 0.33**	**91.87 ± 0.06**	**64.39±0.23**	**86.95 ± 0.05**	**54.05 ±**

Agent	BG3 ($\gamma = 0.98, R = 5$)				
	MENE Regret	NW%	NW$^+$ %	UW %	EF1 %
sonnet-3.7-c-5	25.83±0.93	84.91±0.09	54.41±.20	84.76±0.07	35.24±0.20
sonnet-3.7-c-6	29.57±0.85	86.60±0.13	56.64±0.19	85.88±0.08	32.24±0.21
sonnet-3.7-r-c-0	13.71±0.70	90.61±0.10	52.99±0.20	86.74±0.07	43.34±0.17
gem-2.0-f-c-2	41.49±1.36	85.33±0.15	42.73±0.22	82.56±0.10	25.04±0.21
gem-2.0-f-c-5	55.44±1.32	81.12±0.20	52.33±0.28	81.99±0.13	34.52±0.43
4o-c-4	25.59±0.83	88.79±0.07	57.44±0.21	83.81±0.06	41.53±0.18
4o-c-5	22.69±0.84	88.92±0.14	54.42±0.20	85.10±0.10	37.37±0.21
4o-c-6	7.69±0.56	88.66±0.08	54.84±0.18	84.64±0.06	38.54±0.20
o3-mini-c-0	**4.24 ± 0.45**	**92.08 ± 0.11**	**59.58±0.24**	**87.28 ± 0.09**	**54.62 ± 0.25**

Table 3: Bootstrap means ($\pm$ standard errors) of regret, normalized Nash, NW^+, and utilitarian welfares, and EF1 frequencies, the last 4 expressed as percent points, for our `LLM-only` analysis on BG4–6 and BGS0. Lower values are better for regret and worse for the other metrics. The best value of each metric is in boldface.

Agent	BG4 ($\gamma = 0.9$, $R = 3$)					BG5 ($\gamma = 0.98$, $R = 3$)				
	MENE Regret	NW%	NW^+ %	UW%	EF1%	MENE Regret	NW%	NW^+ %	UW%	EF1%
sonnet-3.7-c-5	26.24 ± 0.61	79.66 ± 0.08	48.56±.21	87.14 ± 0.06	27.59 ± 0.19	28.77 ± 0.73	78.50 ± 0.14	57.05±.28	86.70 ± 0.08	31.05 ± 0.28
sonnet-3.7-c-6	18.34 ± 0.57	79.25 ± 0.07	57.27±.22	86.82 ± 0.07	29.45 ± 0.16	22.64 ± 0.83	77.90 ± 0.14	51.62±.17	86.66 ± 0.09	33.49 ± 0.35
sonnet-3.7-r-c-0	15.39 ± 0.58	82.26 ± 0.07	55.92±.21	**88.70 ± 0.05**	38.38 ± 0.23	17.42 ± 0.61	82.51 ± 0.11	57.52±.21	89.69 ± 0.08	40.44 ± 0.31
gemini-2.0-f-c-2	15.65 ± 0.55	80.61 ± 0.07	51.39±.31	87.15 ± 0.05	32.95 ± 0.27	20.30 ± 0.64	79.97 ± 0.12	48.13±.19	87.76 ± 0.08	34.70 ± 0.27
gemini-2.0-f-c-5	27.89 ± 0.55	78.34 ± 0.11	51.05±.18	86.84 ± 0.07	28.36 ± 0.27	18.17 ± 0.58	75.53 ± 0.21	44.38±.18	86.34 ± 0.12	23.63 ± 0.24
4o-c-4	15.12 ± 0.59	81.51 ± 0.07	48.75±.20	86.61 ± 0.06	31.41 ± 0.25	14.26 ± 0.57	82.96 ± 0.10	56.04±.28	88.31 ± 0.08	34.84 ± 0.35
4o-c-5	16.88 ± 0.52	81.98 ± 0.11	53.56±.27	87.83 ± 0.09	31.35 ± 0.27	27.37 ± 0.66	79.25 ± 0.13	52.62±.17	86.36 ± 0.08	32.13 ± 0.28
4o-c-6	23.18 ± 0.56	81.24 ± 0.08	46.86±.15	87.17 ± 0.07	28.80 ± 0.20	31.05 ± 0.61	79.79 ± 0.11	55.74±.21	85.94 ± 0.07	34.75 ± 0.23
o3-mini-c-0	**0.50 ± 0.10**	**83.59 ± 0.07**	**63.21±.30**	87.45 ± 0.06	**51.57 ± 0.21**	**0.94 ± 0.16**	**85.45 ± 0.10**	**72.04±.49**	**90.01 ± 0.08**	**57.34 ± 0.33**

Agent	BG6 ($\gamma = 0.98$, $R = 5$)					BGS0 (uniform mixture of BG4–BG6)				
	MENE Regret	NW%	NW^+ %	UW%	EF1%	MENE Regret	NW%	NW^+ %	UW%	EF1%
sonnet-3.7-c-5	10.48 ± 0.50	78.29 ± 0.11	54.87±.21	86.83 ± 0.07	30.03 ± 0.27	28.64 ± 0.50	79.85 ± 0.13	53.49±.14	86.94 ± 0.06	34.20 ± 0.28
sonnet-3.7-c-6	21.31 ± 0.58	76.30 ± 0.11	55.90±.20	85.47 ± 0.06	27.95 ± 0.25	26.09 ± 0.54	78.13 ± 0.12	54.16±.13	86.67 ± 0.06	34.15 ± 0.29
sonnet-3.7-r-c-0	16.17 ± 0.65	80.64 ± 0.10	54.32±.22	86.94 ± 0.06	38.35 ± 0.26	22.87 ± 0.50	82.34 ± 0.09	55.71±.15	88.51 ± 0.05	44.18 ± 0.33
gemini-2.0-f-c-2	20.25 ± 0.60	79.80 ± 0.15	54.67±.19	86.33 ± 0.09	32.40 ± 0.22	22.50 ± 0.46	80.74 ± 0.13	54.19±.20	87.04 ± 0.06	38.52 ± 0.28
gemini-2.0-f-c-5	21.80 ± 0.70	73.78 ± 0.11	54.65±.15	83.77 ± 0.07	27.38 ± 0.26	22.48 ± 0.54	77.63 ± 0.18	49.46±.12	86.24 ± 0.07	31.81 ± 0.31
4o-c-4	14.02 ± 0.54	80.31 ± 0.07	54.12±.19	85.54 ± 0.06	31.16 ± 0.23	15.74 ± 0.42	82.08 ± 0.07	53.45±.16	86.72 ± 0.04	38.19 ± 0.36
4o-c-5	16.35 ± 0.55	79.26 ± 0.11	54.21±.17	86.24 ± 0.06	31.33 ± 0.30	22.09 ± 0.40	81.08 ± 0.12	56.09±.17	87.20 ± 0.06	38.04 ± 0.33
4o-c-6	13.68 ± 0.54	80.83 ± 0.12	49.36±.17	87.58 ± 0.07	35.88 ± 0.35	22.72 ± 0.42	81.01 ± 0.11	50.44±.11	86.82 ± 0.06	38.07 ± 0.29
o3-mini-c-0	**1.50 ± 0.19**	**85.04 ± 0.11**	**60.44±.25**	**89.08 ± 0.10**	**51.98 ± 0.25**	**0.12 ± 0.04**	**86.34 ± 0.11**	**68.19±.33**	**89.76 ± 0.07**	**57.40 ± 0.30**

I NFSP training parameters for games in the BGS class

Table 6: NFSP training configuration for BG4 ($\gamma = 0.9$, $R = 3$)

NFSP / training hyper-parameters	
anticipatory parameter η	0.1
ε-greedy exploration (ε)	0.06
batch_size	128
policy_buffer_size	100,000
q_buffer_size	20,000
updates_per_loop	3
policy optimiser	Adam
Q-network optimiser	Adam
policy learning rate	1×10^{-3}
Q-network learning rate	1×10^{-3}
hidden layer size (both nets)	256
policy loss	Cross-Entropy
Q-network loss	Mean-Squared Error
Number of Game Trajectories	10^8

Table 4: Bootstrap means ($\pm$ standard errors) of regret, normalized Nash, NW$^+$, and utilitarian welfares, and EF1 frequencies, the last 4 expressed as percent points, for our `LLM+baselines` analysis on BG1–3. Lower values are better for regret and worse for the other metrics. The best value of each metric is in boldface.

Agent	BG1 ($\gamma = 0.9$, $R = 3$)					BG2 ($\gamma = 0.98$, $R = 3$)			
	MENE Regret	NW%	NW$^+$ %	UW%	EF1%	MENE Regret	NW%	NW$^+$ %	UW%
sonnet-3.7-c-5	40.72 ± 1.33	88.41 ± 0.17	$50.08 \pm .32$	86.77 ± 0.08	33.18 ± 0.20	48.12 ± 1.38	84.93 ± 0.18	$50.85 \pm .27$	84.72 ± 0.07
sonnet-3.7-c-6	26.69 ± 1.33	89.59 ± 0.20	$47.00 \pm .29$	87.31 ± 0.08	32.21 ± 0.20	27.34 ± 1.19	86.22 ± 0.19	$56.07 \pm .24$	85.47 ± 0.08
sonnet-3.7-r-c-0	36.73 ± 0.85	87.19 ± 0.14	$44.31 \pm .27$	84.04 ± 0.07	41.61 ± 0.19	14.21 ± 0.72	87.30 ± 0.15	$53.23 \pm .27$	84.47 ± 0.08
gemini-2.0-f-c-2	37.69 ± 0.88	84.94 ± 0.08	$40.46 \pm .26$	82.84 ± 0.06	34.39 ± 0.20	30.64 ± 0.90	83.45 ± 0.14	$46.20 \pm .23$	82.17 ± 0.08
gemini-2.0-f-c-5	50.76 ± 0.90	84.16 ± 0.13	$41.57 \pm .28$	83.74 ± 0.08	25.72 ± 0.24	32.40 ± 0.92	81.91 ± 0.13	$41.21 \pm .24$	83.85 ± 0.08
4o-c-4	14.65 ± 0.55	91.87 ± 0.09	$47.11 \pm .29$	86.59 ± 0.07	36.27 ± 0.22	19.80 ± 0.76	88.43 ± 0.13	$51.06 \pm .25$	84.44 ± 0.09
4o-c-5	5.91 ± 0.43	92.48 ± 0.11	$48.61 \pm .32$	87.66 ± 0.08	36.51 ± 0.24	9.32 ± 0.62	89.26 ± 0.10	$51.94 \pm .24$	85.42 ± 0.07
4o-c-6	17.77 ± 0.74	$\mathbf{92.78 \pm 0.10}$	$52.07 \pm .33$	$\mathbf{87.72 \pm 0.08}$	37.37 ± 0.26	15.82 ± 0.72	88.95 ± 0.10	$54.73 \pm .30$	85.35 ± 0.08
o3-mini-c-0	$\mathbf{4.37 \pm 0.34}$	92.25 ± 0.10	$\mathbf{54.81 \pm .36}$	86.69 ± 0.07	$\mathbf{50.84 \pm 0.26}$	$\mathbf{4.07 \pm 0.43}$	$\mathbf{90.47 \pm 0.12}$	$\mathbf{60.38 \pm .34}$	$\mathbf{85.68 \pm 0.10}$
soft	106.52 ± 1.81	79.95 ± 0.28	$36.00 \pm .61$	77.13 ± 0.20	21.15 ± 0.43	46.91 ± 1.72	80.23 ± 0.28	$40.81 \pm .53$	79.99 ± 0.16
tough	108.13 ± 2.23	76.11 ± 0.28	$1.62 \pm .05$	79.18 ± 0.18	0.00 ± 0.00	67.82 ± 2.05	63.60 ± 0.19	$1.85 \pm .03$	73.46 ± 0.14
walk	79.65 ± 0.73	72.01 ± 0.00	$0.00 \pm .00$	72.08 ± 0.00	0.00 ± 0.00	94.83 ± 0.71	72.00 ± 0.00	$0.00 \pm .00$	72.08 ± 0.00

Agent	BG3 ($\gamma = 0.98$, $R = 5$)				
	MENE Regret	NW%	NW$^+$ %	UW%	EF1%
sonnet-3.7-c-5	52.47 ± 1.90	83.21 ± 0.25	$54.80 \pm .46$	83.87 ± 0.10	35.47 ± 0.25
sonnet-3.7-c-6	61.69 ± 2.19	85.66 ± 0.26	$54.87 \pm .36$	85.13 ± 0.12	34.79 ± 0.28
sonnet-3.7-r-c-0	26.38 ± 1.35	88.57 ± 0.23	$47.39 \pm .31$	85.36 ± 0.13	42.79 ± 0.28
gemini-2.0-f-c-2	56.86 ± 1.69	84.28 ± 0.18	$41.15 \pm .30$	80.90 ± 0.13	27.81 ± 0.23
gemini-2.0-f-c-5	71.36 ± 2.04	79.17 ± 0.30	$47.28 \pm .39$	81.41 ± 0.16	36.03 ± 0.50
4o-c-4	37.12 ± 1.25	88.16 ± 0.14	$55.82 \pm .37$	83.49 ± 0.11	38.22 ± 0.29
4o-c-5	29.71 ± 1.43	87.87 ± 0.21	$52.08 \pm .35$	84.18 ± 0.16	36.97 ± 0.29
4o-c-6	17.88 ± 1.06	89.05 ± 0.21	$50.43 \pm .36$	85.73 ± 0.18	38.11 ± 0.27
o3-mini-c-0	$\mathbf{3.25 \pm 0.46}$	$\mathbf{92.82 \pm 0.16}$	$\mathbf{57.90 \pm .40}$	$\mathbf{87.71 \pm 0.13}$	$\mathbf{54.68 \pm 0.37}$
soft	33.95 ± 1.61	88.18 ± 0.40	$45.62 \pm .69$	85.69 ± 0.25	32.00 ± 0.42
tough	89.61 ± 2.37	60.63 ± 0.28	$4.94 \pm .08$	71.98 ± 0.15	0.32 ± 0.03
walk	104.13 ± 1.10	72.01 ± 0.00	$0.00 \pm .00$	72.08 ± 0.00	0.00 ± 0.00

Table 7: NFSP training configuration for BG5 ($\gamma = 0.98$, $R = 3$)

NFSP / training hyper-parameters	
anticipatory parameter η	0.1
ε-greedy exploration (ε)	0.06
batch_size	128
policy_buffer_size	1,000,000
q_buffer_size	500,000
updates_per_loop	6
policy optimiser	Adam
Q-network optimiser	Adam
policy learning rate	1×10^{-5}
Q-network learning rate	1×10^{-3}
hidden layer size (both nets)	256
policy loss	Cross-Entropy
Q-network loss	Mean-Squared Error
Number of Game Trajectories	10^8

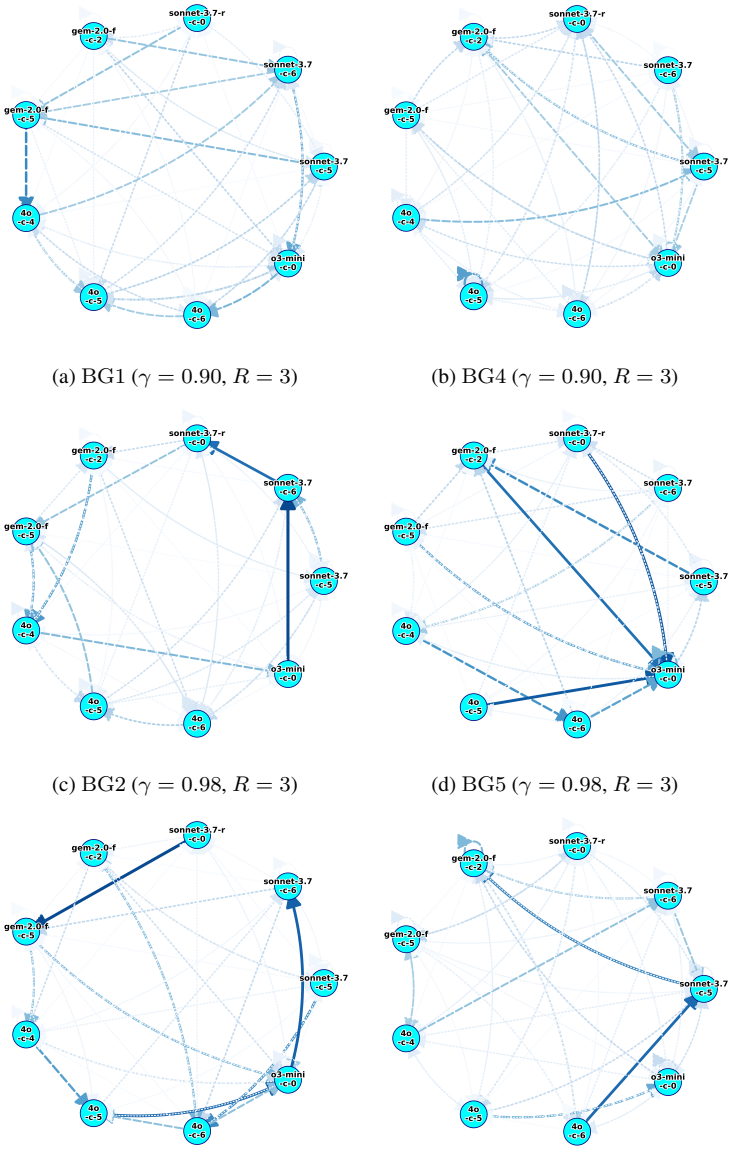
Table 8: NFSP training configuration for BG6 ($\gamma = 0.98$, $R = 3$)

NFSP / training hyper-parameters	
anticipatory parameter η	0.1
ε-greedy exploration (ε)	0.06
batch_size	128
policy_buffer_size	2,000,000
q_buffer_size	1,000,000
updates_per_loop	10
policy optimiser	Adam
Q-network optimiser	Adam
policy learning rate	1×10^{-5}
Q-network learning rate	1×10^{-3}
hidden layer size (both nets)	256
policy loss	Cross-Entropy
Q-network loss	Mean-Squared Error
Number of Game Trajectories	10^8

Table 5: Bootstrap means (± standard errors) of regret, normalized Nash, NW^+, and utilitarian welfares, and EF1 frequencies, the last 4 expressed as percent points, for our `LLM+baselines` analysis on BG4–6. Lower values are better for regret and worse for the other metrics. The best value of each metric is in boldface.

Agent	BG4 ($\gamma = 0.9, R = 3$)					BG5 ($\gamma = 0.98, R = 3$)				
	MENE Regret	NW%	NW^+ %	UW %	EF1 %	MENE Regret	NW%	NW^+ %	UW %	EF1 %
sonnet-3.7-c-5	17.19±.44	78.79±.07	49.43±.16	86.97±.05	29.61±.15	43.27±.81	76.54±.20	55.89±.38	85.22±.10	29.16±
sonnet-3.7-c-6	6.82±.36	77.77±.07	54.04±.20	86.41±.06	30.87±.16	27.12±.80	78.08±.15	54.50±.38	86.32±.10	32.32±
sonnet-3.7-r-c-0	18.08±.54	80.13±.08	50.23±.18	86.68±.06	41.43±.18	21.03±.58	80.62±.12	58.44±.41	87.55±.11	39.21±
gemini-2.0-f-c-2	17.86±.46	78.05±.10	47.94±.20	85.81±.07	35.45±.21	18.93±.57	80.06±.15	50.62±.35	87.67±.10	34.13±
gemini-2.0-f-c-5	24.28±.39	76.94±.20	51.54±.26	86.14±.06	29.38±.19	22.96±.59	75.94±.22	44.09±.29	86.52±.12	25.00±
4o-c-4	17.60±.47	79.92±.07	50.00±.14	85.90±.06	33.74±.21	16.40±.50	82.01±.13	55.80±.39	87.45±.11	34.41±
4o-c-5	14.74±.40	79.97±.09	52.35±.19	86.67±.08	34.65±.22	26.42±.58	75.94±.22	49.34±.31	86.12±.10	31.50±
4o-c-6	11.22±.36	80.81±.09	45.64±.12	87.96±.08	30.27±.16	23.91±.55	80.00±.14	51.20±.33	86.47±.11	31.96±
o3-mini-c-0	1.60±.20	**82.62±.07**	**56.79±.21**	87.56±.05	50.76±.17	**2.36±.23**	**85.65±.14**	**66.83±.55**	**90.05±.11**	**54.79±**
nfsp	13.58±.51	77.91±.08	44.73±.16	84.00±.06	40.57±.23	17.90±.56	82.62±.22	65.05±.36	89.59±.14	52.32±
rnad	**1.17±.16**	80.49±.07	53.70±.20	87.90±.07	**53.72±.20**	4.80±.31	85.03±.17	56.39±.37	90.15±.13	48.27±
aspire	25.83±.50	81.45±.08	43.72±.11	**88.32±.07**	30.73±.09	24.61±.47	84.03±.13	62.49±.36	88.29±.10	46.53±
soft	21.53±.53	70.02±.08	42.98±.19	80.12±.05	24.75±.11	36.69±.69	70.37±.12	46.43±.35	80.47±.06	27.59±
tough	44.77±.53	63.50±.16	9.51±.05	79.09±.06	1.55±.05	49.97±.83	64.34±.18	6.69±.08	77.45±.07	.07±.
walk	37.08±.30	67.97±0.00	0.00±0.00	75.03±0.00	0.00±0.00	60.95±.58	67.98±0.00	0.00±0.00	75.04±0.00	0.00±0

Agent	BG6 ($\gamma = 0.98, R = 5$)				
	MENE Regret	NW%	NW^+ %	UW %	EF1 %
sonnet-3.7-c-5	8.52±.36	78.05±.09	59.34±.26	86.78±.06	31.17±.20
sonnet-3.7-c-6	13.72±.42	76.80±.09	59.26±.26	85.98±.05	29.68±.20
sonnet-3.7-r-c-0	9.30±.36	80.02±.09	54.76±.24	86.87±.07	37.95±.20
gemini-2.0-f-c-2	17.01±.41	79.57±.09	51.43±.19	86.52±.07	32.21±.17
gemini-2.0-f-c-5	26.41±.55	74.77±.11	56.50±.19	84.34±.06	28.97±.19
4o-c-4	7.42±.32	81.23±.08	58.44±.25	86.56±.06	34.81±.25
4o-c-5	12.66±.36	79.54±.08	55.37±.21	86.46±.60	34.18±.22
4o-c-6	13.90±.39	80.10±.08	49.63±.17	86.60±.60	33.33±.24
o3-mini-c-0	**2.13±.23**	**84.05±.08**	61.05±.24	87.86±.07	53.86±.25
nfsp	2.85±.25	82.33±.08	65.79±.36	**87.42±.07**	**55.83±.30**
rnad	15.42±.52	81.25±.12	51.75±.20	86.50±.08	42.94±.27
aspire	11.26±.41	83.85±.08	**67.23±.20**	88.35±.06	46.36±.19
soft	20.52±.50	69.32±.11	46.27±.19	80.13±.04	25.18±.15
tough	36.58±.66	64.51±.13	6.11±.06	77.21±.05	0.18±.03
walk	46.55±.33	67.97±0.00	0.00±0.00	75.04±0.04	0.00±0.00



(a) BG1 ($\gamma = 0.90$, $R = 3$)

(b) BG4 ($\gamma = 0.90$, $R = 3$)

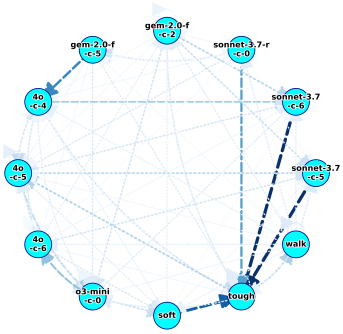
(c) BG2 ($\gamma = 0.98$, $R = 3$)

(d) BG5 ($\gamma = 0.98$, $R = 3$)

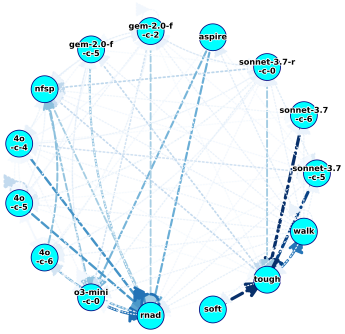
(e) BG3 ($\gamma = 0.98$, $R = 5$)

(f) BG6 ($\gamma = 0.98$, $R = 5$)

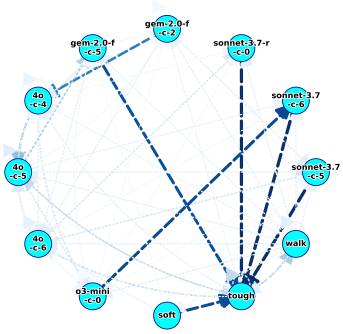
Figure 21: `LLM-only` empirical best-response graphs for BG1–6.



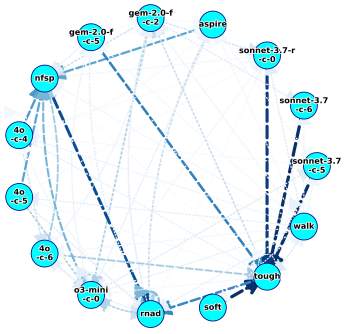
(a) BG1 ($\gamma = 0.90$, $R = 3$)



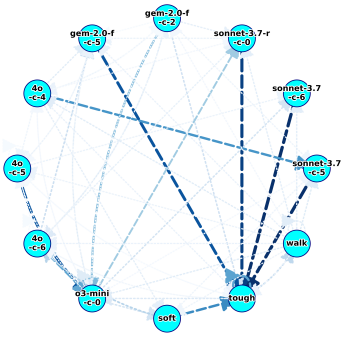
(b) BG4 ($\gamma = 0.90$, $R = 3$)



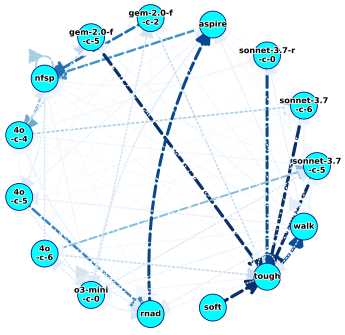
(c) BG2 ($\gamma = 0.98$, $R = 3$)



(d) BG5 ($\gamma = 0.98$, $R = 3$)



(e) BG3 ($\gamma = 0.98$, $R = 5$)



(f) BG6 ($\gamma = 0.98$, $R = 5$)

Figure 22: `LLM+baselines` empirical best-response graphs for BG1–6.

J R-NaD training parameters for games in the BGS class

Table 9: R-NaD training configuration for BG4 ($\gamma = 0.9$, $R = 3$)

R-NaD training hyper-parameters	
trajectory_max	6
policy network hidden layers	[256, 256]
batch_size	256
learning_rate	5×10^{-5}
clip_gradient	1.0
target_network_avg	0.01
training_steps	1,000,000
Adam $(\beta_1, \beta_2, \epsilon)$	$(0.9,\ 0.999,\ 10^{-8})$
entropy_schedule_repeats	[1, 2, 1]
entropy_schedule_size	[5,000, 10,000, 20,000]
η (reward transform)	0.2
NeuRD β	2.0
NeuRD clip	150
V-trace c	1.0
fine-tune from_learner_steps	800,000
policy_threshold	0.03
policy_discretization	32
seed	42

Table 10: R-NaD training configuration for BG5 ($\gamma = 0.98$, $R = 3$)

R-NaD training hyper-parameters	
trajectory_max	6
policy network hidden layers	[256, 256]
batch_size	256
learning_rate	5×10^{-5}
clip_gradient	1.0
target_network_avg	0.01
training_steps	1,000,000
Adam $(\beta_1, \beta_2, \epsilon)$	$(0.9,\ 0.999,\ 10^{-8})$
entropy_schedule_repeats	[1, 2, 1]
entropy_schedule_size	[5,000, 10,000, 20,000]
η (reward transform)	0.2
NeuRD β	2.0
NeuRD clip	150
V-trace c	1.0
fine-tune from_learner_steps	800,000
policy_threshold	0.03
policy_discretization	32
seed	42

Table 11: R-NaD training configuration for BG6 ($\gamma = 0.98$, $R = 5$)

R-NaD training hyper-parameters	
trajectory_max	10
policy network hidden layers	[256, 256]
batch_size	256
learning_rate	5×10^{-5}
clip_gradient	1.0
target_network_avg	0.01
training_steps	1,000,000
Adam $(\beta_1, \beta_2, \epsilon)$	$(0.9,\ 0.999,\ 10^{-8})$
entropy_schedule_repeats	[1, 2, 1]
entropy_schedule_size	[5,000, 10,000, 20,000]
η (reward transform)	0.2
NeuRD β	2.0
NeuRD clip	150
V-trace c	1.0
fine-tune from_learner_steps	800,000
policy_threshold	0.03
policy_discretization	32
seed	42

K Training PPO

Table 12: PPO training configuration for BG4 ($\gamma = 0$, $R = 3$)

PPO training hyper-parameters	
policy architecture	2-layer MLP
hidden size	256
optimizer	Adam
learning rate	1×10^{-3}
PPO epochs per update	4
mini-batch size	512
clipping parameter	0.2
entropy coefficient	0.01
gradient clipping	0.5
number of parallel environments	4096
training iterations	200
minimum episodes per iteration	2000
self-play	Yes
separate policies for players	Yes
maximum episode length	12 steps
policy input	local observation only
advantage baseline	outside option or walk value
minimum completed trajectories	4×10^5

K.1 PPO Description

We train agents using Proximal Policy Optimization (PPO) in self-play. PPO is fully decentralized in our setting: each player's policy is trained from only that player's local bargaining observation, with no centralized value function that observes both players jointly. Separate policies are trained for Player 1 and Player 2, and each policy conditions only on its own local bargaining observation. The policy is represented by a two-layer multilayer perceptron with hidden size 256.

Trajectories are collected from 4096 parallel bargaining environments. The policy is optimized with the PPO clipped surrogate objective, using clipping parameter 0.2 and an entropy regularization coefficient of 0.01. Optimization is performed with Adam at learning rate 10^{-3}. Each training iteration collects at least 2000 completed episodes, and updates are run for 4 PPO epochs with mini-batches of size 512.

Because the bargaining game has terminal payoffs and we use $\gamma = 0$, learning is driven entirely by episode outcomes. The training signal is computed relative to the player's outside option, or walk value, which serves as the baseline for the policy update. Training runs for 200 iterations, corresponding to at least 4×10^5 completed bargaining trajectories.

L Training MAPPO

Table 13: MAPPO training configuration for BG4 ($\gamma = 0$, $R = 3$)

MAPPO training hyper-parameters	
actor architecture	2-layer MLP
critic architecture	centralized 2-layer MLP
actor hidden size	256
critic hidden size	512
optimizer	Adam
learning rate	1×10^{-3}
PPO epochs per update	4
mini-batch size	512
clipping parameter	0.2
entropy coefficient	0.01
gradient clipping	0.5
number of parallel environments	4096
training iterations	200
minimum episodes per iteration	2000
parameter sharing	No
self-play	Yes
maximum episode length	12 steps
critic input	joint observation of both players
actor input	local observation only
advantage baseline	outside option or walk value
minimum completed trajectories	4×10^5

L.1 MAPPO Description

We train agents using Multi-Agent Proximal Policy Optimization (MAPPO), a centralized training and decentralized execution method. Each player acts from its own local bargaining observation, but during training a centralized critic is used that receives the joint state formed from both players' observations. During execution, each player's actor conditions only on its own local bargaining observation. During training, each player is paired with a centralized critic that receives the joint state, formed by concatenating both players' observations. This allows the value function to use information about the full bargaining situation while preserving decentralized policies at test time.

In our bargaining setting, training is performed in self-play with separate policies for Player 1 and Player 2. At each update, trajectories are collected from 4096 parallel bargaining environments. The actor is optimized with the PPO clipped surrogate objective, using clipping parameter 0.2 and an entropy regularization coefficient of 0.01. Both actor and critic are trained with Adam using learning rate 10^{-3}. The actor network uses two hidden layers of size 256, while the centralized critic uses two hidden layers of size 512.

Because the bargaining game has terminal payoffs and we use $\gamma = 0$, learning is driven by episode outcomes rather than long horizon bootstrapping. Each training iteration collects at least 2000 completed episodes, and updates are run for 4 PPO epochs with mini-batches of size 512. Training runs for 200 iterations, corresponding to at least 4×10^5 completed bargaining trajectories.

M Training PSRO

Table 14: PSRO training configuration for BG4 ($\gamma = 0$, $R = 3$)

PSRO training hyper-parameters	
policy architecture	2-layer MLP
hidden size	256
initial population size	1 policy per player
PSRO outer iterations	20
meta-solver	entropy-regularized replicator dynamics
meta-solver iterations	3000
meta-solver entropy weight	0.001
number of parallel environments	4096
episodes per best response	10000
episodes per payoff evaluation	2000
episodes per BR update iteration	3000
PPO epochs per BR update	4
mini-batch size	512
clipping parameter	0.2
entropy coefficient	0.01
gradient clipping	0.5
optimizer	Adam
learning rate	1×10^{-3}
training target	best response to opponent meta-strategy
evaluation metric	NashConv
population update	add 1 best response per player per iteration

M.1 PSRO Description

We train agents using Policy Space Response Oracles (PSRO), a population-based game-theoretic learning procedure. Unlike PPO and MAPPO, which train a single policy for each player in self-play, PSRO maintains a growing population of policies for Player 1 and Player 2. At each outer iteration, the algorithm first estimates an empirical payoff matrix by evaluating every policy pair in the current populations.

The resulting empirical meta-game is then solved for an approximate Nash equilibrium using max entropy nash equilibrium solver. This produces a meta-strategy for each player, that is, a probability distribution over the policies currently in that player's

population. New policies are then trained as best responses to the opponent's meta-strategy rather than to a single fixed opponent.

In our bargaining game, each best response is trained with PPO-style updates using a two-layer MLP policy with hidden size 256. Best response training uses Adam with learning rate 10^{-3}, 4 PPO epochs per update, mini-batch size 512, clipping parameter 0.2, and entropy regularization coefficient 0.01. Training is run with 4096 parallel bargaining environments, and each outer PSRO iteration trains one new best response for each player.

After best response training, the new policies are evaluated against the opponent meta-strategy to compute empirical regret, which measures how much either player can gain by deviating from the current meta-strategy. The trained best responses are then added to the policy populations, and the process repeats. In this way, PSRO constructs a progressively richer empirical game and an increasingly robust mixed strategy over the learned bargaining policies.

N EF1 bargainer

Set up

- Rounds $t \in \{1, \dots, N\}$ with $N \in \{3, 5\}$.
- Discount per round γ^{t-1}, $\gamma \in \{0.9, 0.98, 1\}$.
- Items with quantities $q = (7, 4, 1)$ across $T = 3$ types. Each player has private values v_i drawn per game.
- An allocation assigns items to each player: agent keeps k and opponent receives $q - k$.
- BATNA B is the outside option value, sampled per game.

EF1 Fairness Criterion

An allocation $(k, q - k)$ is *envy-free up to one item* (EF1) for the agent if, letting $m = q - k$ denote the opponent's bundle:

$$v(k) \geq v(m) - \max_{j:m_j>0} v_j$$

That is, the agent does not envy the opponent after hypothetically removing the single most valuable item type from the opponent's bundle. If $v(k) \geq v(m)$ the allocation is envy-free (EF), which implies EF1.

Offer Construction

Feasible set

Enumerate all allocations $k \in \prod_{j=1}^{T} \{0, \dots, q_j\}$ satisfying two conditions:

1. EF1 fairness: $v(k) \geq v(q-k) - \max_{j:(q_j-k_j)>0} v_j$.
2. Individual rationality: $v(k) > B$.

Pareto frontier

From the feasible set, extract the Pareto frontier with respect to the opponent's allocation: an allocation $(k, q-k)$ is on the frontier if no other feasible allocation gives the opponent weakly more of every item and strictly more of at least one. Formally, (k, m) is Pareto-optimal if there is no other feasible (k', m') with $m'_j \geq m_j$ for all j and $m'_j > m_j$ for some j.

Frontier ordering

The frontier is sorted by generosity. The primary key is total items given to the opponent $\sum_j m_j$ (descending). Ties are broken by dominant item type fraction m_j/q_j and then by the fraction vector, ensuring a deterministic ordering across rounds.

Decision Rules

Walk

Walk immediately if:

- No EF1 allocation exists, or
- The best EF1 allocation does not beat the outside option: $\max_{(k,m)\in\text{EF1}} v(k) \leq B$.

Accept

Accept the opponent's offer x if both:

- The allocation is EF1: $v(x) \geq v(q-x) - \max_{j:(q_j-x_j)>0} v_j$.
- The value exceeds the outside option: $v(x) > B$.

Counter-offer

If the opponent's offer is rejected (not EF1 or below BATNA), cycle through the sorted Pareto frontier:

$$\text{offer}_t = \text{frontier}[t \bmod |\text{frontier}|]$$

The opening offer (round 0) is the most generous feasible EF1 allocation. Subsequent rounds rotate through the frontier, emphasizing different item type compositions while maintaining EF1 fairness.

Summary

The EF1 Bargainer is a deterministic, fairness-constrained agent that restricts offers to the Pareto frontier of EF1-feasible allocations. It accepts any opponent offer satisfying EF1 and individual rationality, and cycles through maximally generous offers when countering. The agent uses no opponent modeling, learning, or discounting in its offer logic it applies the same fairness criterion regardless of the round or discount factor. This makes it a principled baseline representing fair division norms in the strategic environment.

O Acceptance % vs. Tough

	Turn level acceptance rate		Game level acceptance rate	
Agent	**vs tough**	**vs others**	**vs tough**	**vs others**
sonnet-3.7-c-5	49.7%	12.3%	78.9%	19.5%
sonnet-3.7-c-6	51.6%	13.8%	85.5%	21.3%
sonnet-3.7-r-c-0	26.4%	10.2%	61.1%	15.8%
gem-2.0-f-c-2	6.4%	5.7%	15.0%	8.9%
gem-2.0-f-c-5	15.3%	9.1%	33.1%	15.5%
4o-c-4	1.9%	1.8%	5.5%	2.7%
4o-c-5	1.5%	3.4%	3.6%	5.1%
4o-c-6	2.6%	3.0%	6.0%	4.5%
o3-mini-c-0	**0.3%**	**0.2%**	**0.6%**	**0.3%**
soft	100.0%	100.0%	100.0%	100.0%
tough	0.0%	0.0%	0.0%	0.0%
walk	0.0%	0.0%	0.0%	0.0%

Table 15: Acceptance of deals with offer value below the BATNA, reported at the turn level and the game level, and split by opponent type. Columns "vs tough" consider only negotiations against `tough`; "vs others" aggregates all other opponents.

This table reports mistaken acceptances: the frequency with which each agent accepts an offer whose value is strictly below its own BATNA. We report this error rate at two granularities. Turn-level acceptance rate measures, across all decision points where the agent had the option to accept, the fraction of turns on which it accepts a below-BATNA offer. Game-level acceptance rate measures, across complete negotiations, the fraction of games in which the agent ever makes such an acceptance (hence these rates are typically higher than turn-level rates). Results are split by opponent type: vs tough restricts to negotiations against the rigid `tough` heuristic, while "vs others" aggregates all remaining opponents.

P Opposition Levels

To better understand bargaining dynamics within our game, we compute the opposition levels for both BGS and BGL in Figs. 23 and 24. Opposition level captures the degree to which the two agents can simultaneously achieve high utility under the same terminal outcome. Following the definition used in NegMAS, we measure this by considering the achievable utility points in normalized utility space and computing the distance of the closest feasible point to the ideal point $(1, 1)$.

Concretely, let $\mathcal{O}$ be the set of feasible terminal outcomes (allocations), including the disagreement/BATNA as a terminal outcome, and let $u_i(o)$ denote player i's utility for the outcome o. Let $U_i^{\max} := \max_{o \in \mathcal{O}} u_i(o)$ be the maximum attainable utility for player i over $\mathcal{O}$. For each outcome o, we compute the normalized utilities $\tilde{u}_i(o) = u_i(o)/U_i^{\max}$ and define its distance to the ideal point as

$$d(o) \;=\; \sqrt{\sum_{i \in \{1,2\}} \left(1 - \tilde{u}_i(o)\right)^2} \;=\; \sqrt{\left(1 - \frac{u_1(o)}{U_1^{\max}}\right)^2 + \left(1 - \frac{u_2(o)}{U_2^{\max}}\right)^2}.$$

The opposition level of the instance is then

$$\mathrm{Opp} \;=\; \min_{o \in \mathcal{O}} d(o).$$

Intuitively, $\mathrm{Opp} = 0$ when there exists an outcome that simultaneously achieves both players' maximum utilities, while larger values indicate stronger opposition because even the closest feasible point remains far from the ideal $(1, 1)$ in normalized utility space.

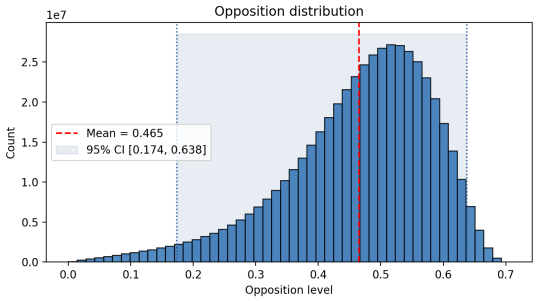


Figure 23: Opposition levels for the BGL setting of games.

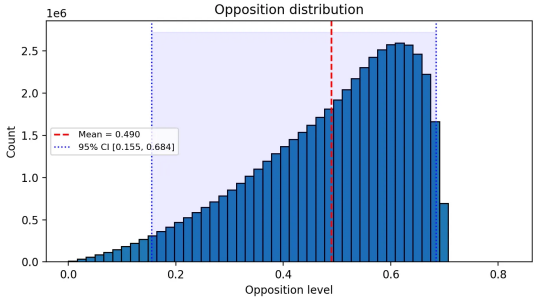


Figure 24: Opposition levels for the BGS setting of games.

Q Provider Level Analysis

Table 16: Equilibrium weighted mean value given (mean $\pm$ SE, $n = 1000$).

Family	**Mean $\pm$ SE**
Anthropic	228.3 ± 0.76
Gemini	213.2 ± 1.94
OpenAI	222.4 ± 0.87

Table 17: Empirical frequency with which the row provider is a best response to the column provider.

	Anthropic	**Gemini**	**OpenAI**
Anthropic	0.04	0.06	0.44
Gemini	0.73	0.00	0.00
OpenAI	0.23	0.94	0.56

The top table 16 gives the amount each model provider gives in equilibrium in the provider level meta-game. The table below 17 gives the empirical frequencies each provider is the best response to another.

Q.1 Description of Provider Level Meta-Game

To construct the provider-level meta-game, we treat each LLM provider as a player. Each provider's strategy set consists of the LLM-based meta-strategies in our library that are supplied by that provider.

We first group strategies by provider. For each ordered pair of providers (A, B), we form the induced two-player subgame between A's strategies $\mathcal{S}_A$ and B's strategies $\mathcal{S}_B$. Let $R \in \mathbb{R}^{|\mathcal{S}_A| \times |\mathcal{S}_B|}$ denote the corresponding payoff matrix for the row player, where each entry R_{ij} is the expected utility achieved by A's i-th strategy when negotiating against B's j-th strategy (estimated from cross-play).

We solve this subgame using the Lemke–Howson algorithm to obtain equilibrium mixtures $\sigma \in \Delta(\mathcal{S}_A)$ and $\tau \in \Delta(\mathcal{S}_B)$. The row provider's equilibrium value for this matchup,

$$v(A, B) \;=\; \sigma^{\top} R \tau,$$

defines the (A, B) entry of the provider-level payoff matrix V. Repeating this procedure for all ordered pairs yields a square provider-level game whose size is the number of providers. We then analyze this provider-level game using the same best response graph pipeline as at the agent level, interpreting directed edges as empirical best responses between providers.

R BGL0

Agent	BGL0 (LLM-only)				
	MENE Regret	NW%	NW$^+$ %	UW %	EF1 %
sonnet-3.7-c-5	33.24±0.77	87.95±0.05	53.33±0.10	85.87±0.05	36.35±0.13
sonnet-3.7-c-6	23.37±0.49	89.61±0.09	55.94±0.11	86.99±0.06	36.29±0.18
sonnet-3.7-r-c-0	20.39±0.61	89.80±0.05	50.49±0.11	85.58±0.04	43.86±0.14
gem-2.0-f-c-2	33.61±0.73	85.94±0.10	45.34±0.13	82.65±0.07	32.68±0.16
gem-2.0-f-c-5	42.62±0.69	85.66±0.10	47.61±0.14	84.42±0.07	35.80±0.35
4o-c-4	20.65±0.51	90.77±0.06	52.77±0.13	85.53±0.05	41.59±0.19
4o-c-5	7.33±0.36	91.99±0.07	54.93±0.14	87.08±0.05	39.62±0.20
4o-c-6	10.46±0.49	90.43±0.04	56.47±0.11	85.81±0.03	40.70±0.20
o3-mini-c-0	**2.12±0.21**	**92.37±0.06**	**63.08±0.18**	**87.16±0.05**	**55.96±0.21**
Agent	BGL0 (LLM+baselines)				
	MENE Regret	NW%	NW$^+$ %	UW %	EF1 %
sonnet-3.7-c-5	42.84±1.32	86.48±.18	53.11±.27	85.08±.07	35.17±.22
sonnet-3.7-c-6	31.79±1.35	88.56±.21	55.51±.29	86.58±.09	35.76±.24
sonnet-3.7-r-c-0	23.28±.65	88.75±.16	50.17±.26	85.00±.07	43.03±.24
gem-2.0-f-c-2	36.14±.74	85.23±.11	44.70±.25	82.17±.07	32.82±.20
gem-2.0-f-c-5	48.14±.92	84.68±.16	46.29±.27	83.80±.09	35.13±.37
4o-c-4	24.08±.56	90.22±.11	51.82±.28	85.21±.07	40.17±.28
4o-c-5	9.16±.50	91.26±.12	53.94±.29	86.49±.09	38.42±.27
4o-c-6	12.25±.56	90.20±.10	54.85±.29	85.75±.07	39.13±.26
o3-mini-c-0	**1.70±.20**	**92.14±.11**	**61.89±.36**	**86.98±.08**	**54.27±.32**
soft	51.58±1.36	85.41±.27	36.94±.34	82.83±.16	24.72±.34
tough	117.33±1.69	68.36±.17	2.66±.04	72.65±.10	0.03±.00
walk	93.54±.67	72.01±0.00	0.00±0.00	72.09±0.00	0.00±.00

Table 18: Bootstrap means (± standard errors) of regret, normalized Nash and utilitarian welfares, and EF1 frequencies, the last 3 expressed as percent points, for our LLM-only and LLM+baselines analyses on BGL0.